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SUPPORTING BPMN EXTENSIONS PROPOSALS THROUGHT A PROCESS

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Qualification proposal submitted to the Academic Master's Degree in Computing of the Federal University of Ceara (UFC) as a partial requirement for the dissertation defense.

Advisor: Prof. Dr. Enyo José Tavares Gonçalves

Co-advisor: Prof. Dr. João Baptista da Silva Araujo Junior

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ABSTRACT

This study addresses the development and application of extensions to the *Business Process Model and Notation* (BPMN) notation, with the aim of improving the processes for creating, managing and disseminating these extensions. The research began with a *Systematic Literature Review* (SLR), whose purpose was to identify the main approaches and challenges related to the extension of BPMN, enabling an in-depth analysis of the gaps and opportunities that exist in this field. The results of this analysis indicated that many of the existing extensions lack clear standardization and complete formal definitions, which limits their adoption and practical applicability. In order to overcome these limitations, a catalog was developed that facilitates the modeling of extensions, providing easy access and promoting the reproducibility of solutions. The main proposal of this work involves the application of *Large Scale Language Models* (LLMs) in the extension process, in order to optimize the creation and adaptation of new solutions. In addition, an extension process was structured for BPMN, inspired by previously established methodologies such as (GONCALVES *et al.*, 2020) and (BRAUN, 2016), and which incorporates the use of LLMs to provide support in the definition and implementation of new extensions. This interactive process follows the standards established by the *Object Management Group* (OMG), with the aim of simplifying the definition of extensions, improving the clarity of their descriptions and encouraging the reuse of pre-existing solutions. The application of the proposed methodologies, combined with the use of the LLMs, represents a significant advance in the way the extensions to the BPMN can be developed and applied, promoting greater standardization, flexibility and effectiveness in the business process modelling process.

Keywords: BPMN. Extensions. Catalog. LLMs. Process Modeling.

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LIST OF ABBREVIATIONS AND ACRONYMS

BPMN	<i>Business Process Model and Notation</i>
SLR	<i>Systematic Literature Review</i>
LLMs	<i>Large Scale Language Models</i>
OMG	<i>Object Management Group</i>
BPM	<i>Business Process Management</i>
UML	<i>Unified Modeling Language</i>
KAOS	<i>Knowledge Acquisition in Automated Specification</i>
ML	<i>Modeling Language</i>
PRISE	<i>A Process to Support iStar Extensions</i>

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1 INTRODUCTION

BPM is a discipline focused on improving the efficiency and effectiveness of organizational operations, through the analysis, modeling, execution, monitoring and optimization of business processes (HILL *et al.*, 2008). With the growing complexity of corporate environments - driven by the diversity of modeling languages, standards and systems - BPM has established itself as a strategic area, both in academia and industry (KO *et al.*, 2009).

Among the various types of standards related to BPM, graphic standards stand out for facilitating communication between stakeholders, offering intuitive visual representations of workflows. In this scenario, BPMN stands out as the main graphical standard for modeling business processes (KO *et al.*, 2009). The notation offers a standardized visual language that enables analysts and developers to represent and share process information in a clear and accessible way.

BPMN(WHITE; MIERS, 2008) is a diagrammatic modeling language widely used for representing and managing business processes. It consists of a standardized set of constructs that facilitates the visual representation of processes, contributing significantly to understanding by users. The idea of construct refers to a fundamental unit in modeling, which allows developers to represent complex aspects of systems under development in a clear and understandable way (BRAMBILLA *et al.*, 2017). In this paper, we use the term construct to refer to the elements of the concrete syntax of BPMN. Thus, by using the terms construct and constructs, we are aligned with the definition proposed by Brambilla (BRAMBILLA *et al.*, 2017).

BPMN aims to increase the competitiveness of organizations by optimizing the productivity of their employees and improving the definition and organization of the company's internal processes (OCA *et al.*, 2015). The effective integration of technology plays a central role, reducing human error, speeding up execution and providing valuable insights for strategic management, as well as enabling the creation and execution of workflows that integrate business and operational processes.

BPMN is defined and maintained by the OMG and specified as an ISO standard (ISO/IEC19510:2013)¹. Version 2.0 consolidated the best practices of previous notations into a single standard, promoting integration and clarity in process modeling. (MALDONADO *et al.*, 2016) points out that the language is becoming a standard in business process modeling, arguing that it offers the advantages of a graphical language, as well as simplicity, ISO standardization

¹ <http://www.omg.org/spec/bpmn/2.0/>

and support for execution processes. Various tools were proposed to support process modeling with BPMN, such as BPMN.io², Activiti³ and Bizagi⁴.

Several modeling languages, such as *Unified Modeling Language* (UML) ((OMG), 2017), iStar (YU, 1995) and *Knowledge Acquisition in Automated Specification* (KAOS) (DARDENNE *et al.*, 1993), have made an effort to identify their existing extensions in order to support the community and provide a comprehensive overview of the work done to extend these languages, as presented by (SILVA *et al.*, 2018a), (GONÇALVES *et al.*, 2018) and (GONÇALVES; MONTE, 2024). The practice of extending modeling languages, in general, is usually motivated by two main factors: the improvement of existing elements and the adaptation of the language to meet the needs of specific domains.

The first approach focuses on improving aspects of the language that are considered valuable, while the second seeks to adapt it to particular contexts, such as areas of study with unique requirements. These issues highlight the need to incorporate new domain-specific elements into the predefined components of BPMN, with the aim of meeting the particularities of sectors such as Healthcare, Internet of Things (IoT) and Quantum Computing. It is essential to emphasize that when extending a language, the new version must maintain compliance with the established standards, preserving the integrity of the original elements.

In addition, the extensions make the language more accessible and increase its chances of adoption, favoring its acceptance in both academia and industry. Extensions not only offer the flexibility needed to meet specific demands, but also take advantage of the benefits of the original language, such as standardization and tool support, thus optimizing the development and application of complex business models, as discussed in (BRAUN; ESSWEIN, 2014b).

Due to its widespread use in business, BPMN has also been adjusted to better represent specific domains, as well as improvements, additions and modifications to existing specifications. The motivation to extend a language usually arises from the need to improve or fill gaps existing in specific contexts. To overcome this limitation and expand BPMN notations, the OMG has introduced an extension mechanism that allows users to integrate new elements and provide valid extensions to BPMN((OMG), 2011).

Several previous studies have analyzed BPMN extensions, including those conducted by (BRAUN; ESSWEIN, 2014a) and (ZAROOUR *et al.*, 2020). The review conducted by

² <https://BPMN.io/>

³ <https://www.activiti.org/>

⁴ <https://www.bizagi.com/pt/plataforma/modeler>

(BRAUN; ESSWEIN, 2014a) was a pioneer, identified in our search, in the analysis of BPMN extensions, compiling a total of 30 papers published between 2007 and 2014. In 2020, (ZAROOUR *et al.*, 2020) broadened this scope by reviewing 49 papers covering the period from 2014 to 2018. These studies highlight the need to further explore BPMN extensions.

In this context, we conducted an SLR focusing on the extensions proposed for BPMN in the period 2019 to 2024. The main objective was to identify and characterize these extensions, understand their contexts of application and impacts, and raise gaps and challenges not yet addressed in previous reviews. Unlike previous studies, our investigation stands out for examining not only the existence of the extensions, but also the conceptual clarity of the constructs introduced, the definition of their contexts of use and adherence to the original structure of the language.

The SLR resulted in the identification of 80 studies proposing extensions to the BPMN. The analysis revealed that 95.6% of the proposals do not comply with the language's original syntax, and 54.4% do not offer effective support for modeling the new elements. In addition, it was found that 53.2% do not provide formal definitions for the constructs introduced, 20.3% present only partial descriptions, and only 26.6% provide complete definitions. These results highlight a critical gap in the formalization and documentation of extensions, which can compromise their reusability, interoperability and adoption in practical environments.

In response to these challenges, we developed an online catalog of BPMN extensions to support researchers and practitioners in formalizing, registering and consulting these proposals. The catalog was fed with data extracted from the SLR and submitted for evaluation by potential users. The results showed a strong intention to use the tool, although its actual use has not yet been consolidated. This discrepancy suggests the presence of external barriers, such as a lack of institutional policies, infrastructure limitations or time constraints for adoption. Despite this, users' perception of the usefulness and usability of the catalog was largely positive.

The analysis showed a growing academic interest in the subject of BPMN extensions, but also revealed the persistence of methodological challenges, especially the absence of clear and well-defined guidelines for the process of creation and formalization of these extensions. In addition, significant opportunities were identified for the evolution of this process, especially regarding the integration with tools based on LLMs, which can provide interactive, automated and contextual support to the development of extensions.

It was observed that, so far, none of the extensions analyzed in SLR explored the

use of LLMs as facilitators in the BPMN extension process. Therefore, this work also seeks to investigate how LLMs can contribute to this process, promoting greater dynamism, intelligent assistance and alignment with modern software engineering practices.

In this context, this research proposes to improve the BPMN extension process, based on the PRISE (GONCALVES *et al.*, 2020) approach and the methodological recommendations proposed by (BRAUN; SCHLIETER, 2014). The objective is to establish a more systematic process, guided by evidence obtained in SLR, aligned with the practices observed in the extension catalog and enriched with the support offered by tools based on LLMs. The proposal aims to make extension development more structured and replicable.

1.1 Research questions

Based on the discussion on the previous section, we formulated the following research questions:

- How have BPMN extensions been proposed and applied according to the scientific literature?
- How to organize and make these extensions available in an accessible, reusable and systematic way for the interested community?
- How can LLMs support the process of defining and documenting BPMN extensions?
- How can a systematic process be defined to guide the creation of new BPMN extensions, based on good practices and the use of LLMs?

1.2 Research Objectives

This work aims to propose a systematic approach to support the BPMN language extension process, based on a literature review, the construction of an existing catalog of extensions and the investigation of the potential use of LLMs as a support tool.

The specific objectives of this work are defined as follows:

- Conduct a SLR on BPMN extensions;
- Build and make available an online catalog with the extensions identified, organized and categorized according to defined criteria;
- Investigate and demonstrate how LLMs can support the process of creating, documenting and applying BPMN extensions;

- Propose a systematic process for the extension of BPMN, incorporating the evidence obtained in SLR, the practices observed in the catalogue and the support offered by LLMs;

1.3 Organization of the document

This dissertation is organized as follows: Chapter 2 presents the background that foundations the work. Chapter 3 discusses the related works, offering a comparative analysis with approaches identified in the literature. Chapter 4 describes the SLR conducted on the extensions to the BPMN language. Chapter 5 presents the online catalog developed as a structured repository of these extensions. The methodology adopted for this dissertation is detailed in Chapter 6. Finally, Chapter 7 presents the final considerations and points to directions for future research.

2 BACKGROUND

This chapter presents the theoretical background to this dissertation, which deals with extensions to the BPMN language. To this end, the fundamental concepts related to the BPMN language are covered, with an emphasis on version 2.0, as well as an overview of *Modeling Languages* (MLs) and its extension mechanisms. The theory related to the development and extension of MLs is also discussed, which is essential for understanding the main stages involved in this process, as described in the specialized literature. Section 2.1 explores the fundamentals of MLs; Section 2.2 discusses the principles of BPM and BPMN. Finally, considering the objectives of this dissertation, Section 2.3 introduces the concepts related to LLMs and Prompts Engineering.

2.1 Modeling Languages

As pointed out by (BRAMBILLA *et al.*, 2017), ML are powerful tools that enable designers to transcribe and conceptualize reality in an explicit and structured way, whether through textual or graphical representations. These languages are made up of three fundamental components:

- **Abstract Syntax:** Describes the fundamental structure of the language, establishing how the elements interrelate, regardless of their visual or textual representations.
- **Concrete Syntax:** Refers to the visual or textual representations of the language, addressing both the graphic and coding aspects that make the language accessible and manipulable by computer tools.
- **Semantics:** Defines the meaning of language elements, specifying the possible relationships between them, according to a pre-established standard, and ensures that interpretations are consistent and correct.

According to (KELLY; TOLVANEN, 2008), the constructs of a ML must be formalized, and more precisely specified by defining its metamodel. This formalization reinforces the importance of presenting the metamodel whenever the language is extended, since it defines the conceptual structure of the language, establishing the permitted elements, their properties and the relationships between them. In addition, it is essential that the language includes domain rules, restrictions and consistency requirements, which ensure the robustness and applicability of the models generated.

Metamodeling, in this context, refers to the process of modeling the language itself, mapping the concepts of the application domain to the elements that make up the language, including their functions and interconnections. Concrete syntax, in turn, consists of the set of visual and textual elements - such as words and graphic symbols - that enable the construction of understandable models and diagrams, as highlighted by (HE *et al.*, 2007).

To ensure that a ML is effective and usable by software designers, it is necessary to define not only the graphic and textual elements that will represent the models, but also how these elements are organized and interact in the concrete syntax, from the base defined by the abstract syntax (BRAMBILLA *et al.*, 2017). In this context, France *et al.* emphasize that modeling tools play a crucial role in reducing the accidental complexity associated with the use of these languages, facilitating implementation and efficient use. The construction of these tools involves the representation of both abstract and concrete syntax, using specialized frameworks that allow models to be created in a more intuitive and effective way.

2.2 Overview of BPM and BPMN Concepts

BPM is a field focused on improving the efficiency and effectiveness of organizational operations through the analysis, modeling and management of business processes. In recent years, the growing variety of ML, standards and software systems related to BPM has created a complexity scenario (HILL *et al.*, 2008).

Given this diversity of approaches, it has become essential to establish a structured classification of existing BPM standards (KO *et al.*, 2009). These standards can be grouped into four main categories:

- **Execution Patterns:** Responsible for implementing and automating business processes, allowing them to be operated on different software platforms.
- **Exchange Standards:** Enable communication between different BPM systems, guaranteeing the consistency and interoperability of the information exchanged.
- **Graphical Standards:** They define visual notations for modeling processes, facilitating their understanding by stakeholders through diagrammatic representations.
- **Diagnostic Standards:** Focused on the analysis and monitoring of running processes, helping to identify bottlenecks and optimize workflows.

In this context, the BPMN plays a essential role as one of the main graphic standards within Business Process Management. The BPMN notation offers a clear and accessible visual

representation of workflows, allowing both business analysts and developers to understand and communicate the structure and functioning of organizational processes effectively. In this way, BPMN is key in the BPM ecosystem, promoting efficient communication and enabling detailed analysis for the continuous improvement of business processes (KO *et al.*, 2009).

BPMN is a widely used ML for representing business processes, and is recognized in both academia and industry (CHINOSI; TROMBETTA, 2012). In 2004, the first version of the BPMN was presented and became widely adopted in academia and industry. BPMN 2.0, released in 2011, introduced a standardized metamodel and a diagram exchange format, allowing models to be exchanged between tools from different vendors. The notation was also extended to include inter-organizational interactions and new elements to model more complex processes, such as sub-processes (AAGESEN; KROGSTIE, 2015).

In this subsection, we focus on BPMN version 2.0, which is the most recent and significant, introducing new features and modifying the properties of the elements as established in previous versions (CHINOSI; TROMBETTA, 2012). BPMN is standardized by the glsOMG and specified as an ISO standard. The recent formalization addresses the execution semantics of all BPMN elements, establishes an extensibility mechanism and resolves inconsistencies and ambiguities, among other significant advances ((OMG), 2011).

BPMN was developed to make it easier for analysts, developers and managers to understand business processes. It aims to eliminate the need for manual translations between process models and their implementations, creating a bridge between visual notation and execution languages. BPMN allows business processes to be modeled at three levels: descriptive, analytical and executable. The descriptive level is the most widely used, while the analytical and executable levels are used for detailed analysis and to implement models that drive process engines (AAGESEN; KROGSTIE, 2015).

The elements of BPMN are organized into five main categories: flow objects, data, connection objects, pools and artifacts. Figure 2.1 illustrates the main components of BPMN 2.0. This language is widely valued for its semantic robustness and expressiveness, which makes models easier to interpret and reduces the risk of misunderstandings. In addition, BPMN is supported by various modeling tools and plugins integrated into some of these platforms, such as BPMN.io, Activiti and Bizagi. These tools enable the efficient creation and conversion of BPMN models into executable languages, facilitating the automation and implementation of business processes (BRAUN *et al.*, 2016).

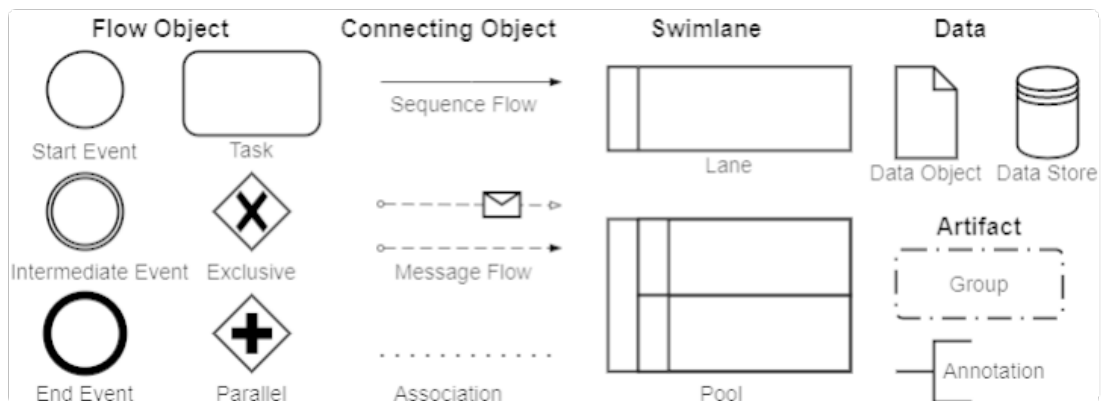


Figure 2.1 – Basic elements of BPMN 2.0

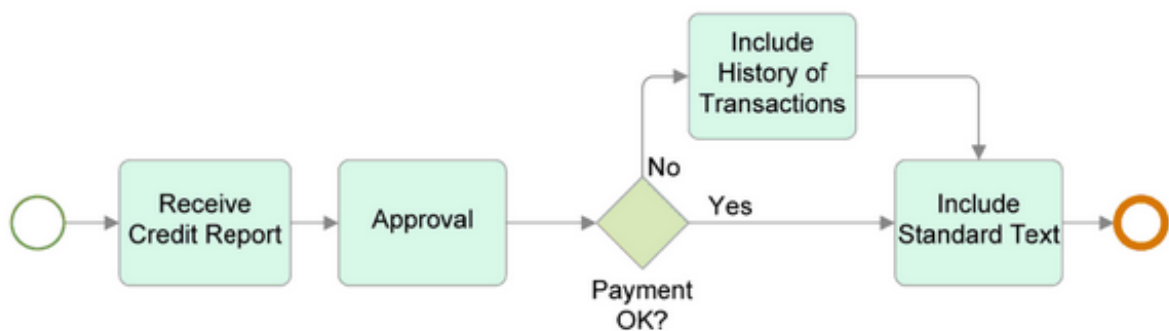


Figure 2.2 – Example of Process Modeling with BPMN

As an example of modeling with BPMN (WHITE; MIERS, 2008), we present the process illustrated in Figure 2.2. This process depicts a typical scenario in which organizations face challenges related to market competition, such as the frequent launch of new products by competitors, growing customer demand for shorter delivery times and more affordable prices, as well as the need to continuously adapt to regulatory changes.

Figure 2.3 shows a detailed view of each element of the process, highlighting the main components of the BPMN notation used in the modeling, such as tasks, events, gateways and associated artifacts.

2.2.1 BPMN Extension Mechanism

According to OMG ((OMG), 2011), each BPMN element that is a subclass of the BPMN *BaseElement* can be extended with additional attributes. The *ExtensionDefinition* groups new attributes under a new concept name and can be created independently of any BPMN definition. However, to use this metaclass to represent an extension, it must be associated with the

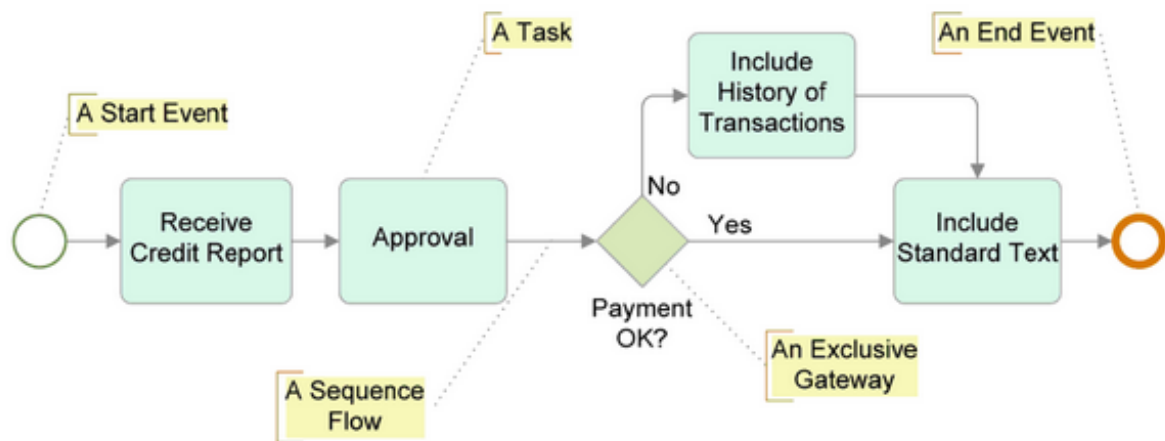


Figure 2.3 – Presentation of the BPMN Notation Components in the Process Model Presented

Extension metaclass, which links an *ExtensionDefinition* and its attributes to the definition of a specific BPMN model. An *ExtensionDefinition* consists of several *ExtensionAttributeDefinitions*, which define the list of new attributes that can be attached to any BPMN element. The values of the new attributes are stored in the *ExtensionAttributeValue* metaclass. The *mustUnderstand* attribute indicates whether the semantics defined by the extension definition must be understood in order to process the BPMN model correctly. The default value is false. As for the *isReference* attribute, it indicates whether the attribute value will be referenced or contained.

The second format for representing valid BPMN extensions is a set of domain-specific elements represented in XML Schema, which is saved in a BPMN file (i.e. a file with a BPMN extension). The XML Schema representation specifies the interchange format for BPMN models and can support the definition of complex extensions that can be processed by BPMN tools. However, the MOF-based extension mechanism has limited capabilities. For example, it does not define the type structures of the new attributes (PILLAT *et al.*, 2015).

In (STROPPI *et al.*, 2011) it was considered that the weak point in the BPMN extension mechanism is the lack of methodological guides for developing BPMN extensions. To fill this gap, they propose a procedure based on Model Driven Architecture (MDA) for the methodical development of valid BPMN extensions. This method consists of four main steps:

- Design of a Conceptual Domain Model Extension (CDME) using UML.
- Definition of a BPMN+X model that describes an extension in terms of the BPMN extension mechanism. In this step, the elements of the CDME are typified as 'BPMN concept' or 'extension concept' using UML stereotypes.
- Transformation of the BPMN+X model into an XML Schema Extension Definition Model.
- Transformation of the XML Schema Extension Definition Model into an XML Schema

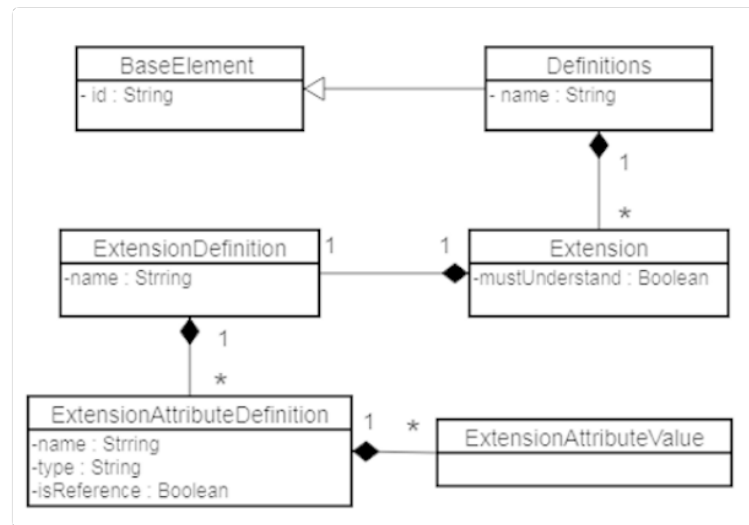


Figure 2.4 – Standard representation of BPMN extension in class diagram

Extension Definition Document.

In contrast, the (STROPPI *et al.*, 2011) method was criticized by (BRAUN *et al.*, 2016) for its lack of analysis of domain requirements. Furthermore, it does not consider whether a domain concept needs to be integrated as a new element or whether BPMN is semantically sufficient to support it. Therefore, (BRAUN *et al.*, 2016) extended the (STROPPI *et al.*, 2011) method with an in-depth analysis of the domain requirements and a comparison of the domain-specific concepts with the basic BPMN concepts.

2.2.2 Extending with OMG Standards

The BPMN metamodel was developed with extensibility in mind, allowing adopters to expand the language while maintaining compatibility. A set of extension elements makes it possible to add attributes and elements to the BPMN standards. It is essential that the standard elements remain unchanged to ensure understanding by other adopters; thus, the extension attributes must not contradict the semantics of any BPMN 2.0 element. This approach is crucial to guaranteeing the validity of the models.

Extensions should also be represented using XML Schema, as shown below:

```

<xsd:element name="extension" type="tExtension"/>
<xsd:complexType name="tExtension">
  <xsd:sequence>
    <xsd:element ref="documentation" minOccurs="0" maxOccurs="unbounded"/>
  </xsd:sequence>

```

```

<xsd:attribute name='definition' type='xsd:QName'/'>
<xsd:attribute name='mustUnderstand' type='xsd:boolean' use='optional'/'>
</xsd:complexType>

```

Another pattern to follow is the representation illustrated in Figure 2.4. The BPMN extension is made up of four main elements that should be presented as follows:

- **Extension:** The Extension element links or imports an ExtensionDefinition and its attributes to a BPMN model definition.
- **ExtensionDefinition:** To be used in a BPMN model it must be linked to an element called Extension, which connects the ExtensionDefinition to a specific BPMN model. This Extension element resides within the BPMN Definitions element, allowing it to be associated with any BPMN element that uses this ExtensionDefinition. Each BPMN element that is a subclass of the BPMN BaseElement can be extended with additional attributes by associating it with an ExtensionDefinition.
- **ExtensionAttributeDefinition:** This element defines a list of attributes that can be attached to any BPMN element. The list specifies the name and type of the new attribute.
- **ExtensionAttributeValue:** The ExtensionAttributeValue element contains the value of the attribute, of type Element.

As an example of a BPMN extension, we present an adaptation of the activity and process elements proposed by (OUARHIM *et al.*, 2020). This extension provides a more accurate representation for the specific domain of real-time manipulation, showing how the adaptation of the elements in the constructs reflects the need for greater specificity in the area. To illustrate the proposal, Figure 2.5 shows a model exemplifying the extension, highlighting the extended Real-Time Activity element, which represents the patient identification process, highlighting the need for real-time components to ensure speed and accuracy in collecting information and assessing symptoms.

2.3 Large Scale Language Models

LLMs are a subset of Artificial Intelligence tools designed to process, understand and generate text in a contextualized way. The accelerated advancement of these models in recent years has significantly transformed the field of Natural Language Processing (NLP),

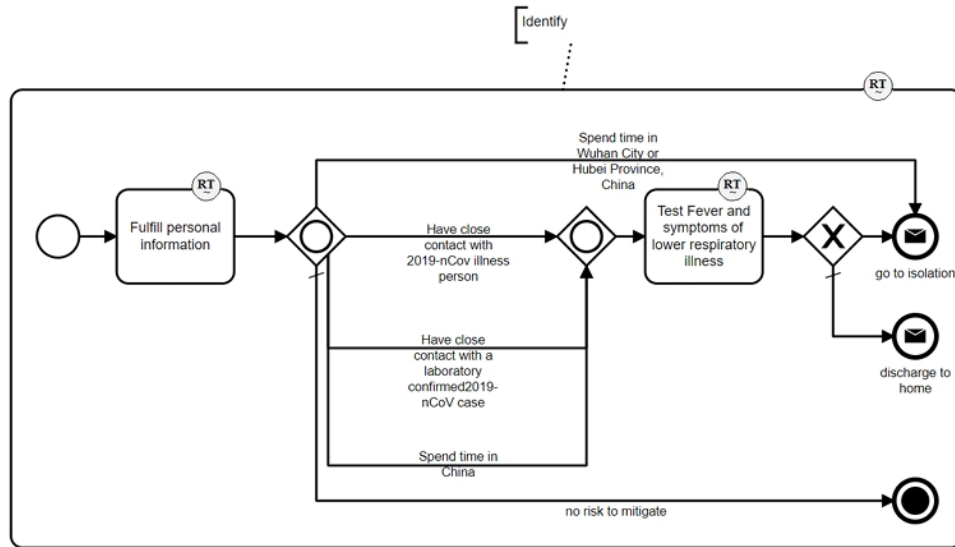


Figure 2.5 – Example model using an extension(OUARHIM *et al.*, 2020)

especially after the public release of ChatGPT¹ by OpenAI in November 2022 (YANG *et al.*, 2024). These models have demonstrated impressive capabilities in tasks such as answering questions, summarizing, translating and paraphrasing texts, often matching human performance (BARI, 2024).

This proficiency is the result of training on large volumes of textual data from various sources, which allows LLMs to generate contextually relevant answers in multiple languages and domains (BROWN *et al.*, 2020). In general, language models refer to systems capable of predicting tokens - whether letters, words or sentences - based on their sequential or adjacent context (BENDER; KOLLER, 2020).

According to (ZHAO *et al.*, 2023), LLMs represent a significant evolution in the development of language models. While the first models were based on static representations, the introduction of the Transformer architecture enabled a scalable leap in their learning capacity. Large-scale language models have hundreds of billions of parameters, adjusted based on massive data sets, and have led significant advances in NLP tasks (FENG *et al.*, 2021).

According to (YANG *et al.*, 2024), current LLMs can be classified into three main architectural categories:

- **BERT-type models (Encoder-only or Encoder-Decoder):** use bidirectional attention mechanisms, in which all the tokens in the input can interact with each other. They are trained with the *Masked Language Modeling* (MLM) technique, where certain words are masked and the model must predict these gaps based on the context. This approach is

¹ <https://chatgpt.com/>

effective for text comprehension tasks such as classification and named entity recognition. Notable examples include BERT and RoBERTa.

- **GPT-type models (Decoder-only):** use causal attention, in which each token can only access the previous tokens in the sequence. They are trained with an autoregressive objective, i.e. they predict the next token based on the previous ones. These models have gained prominence in text generation, machine translation and question answering tasks. Among the main representatives are GPT-3, OPT, PaLM and BLOOM.
- **Hybrid Models (Encoder-Decoder with combined objectives):** integrate features of the previous models. The input is generally divided into two parts: the first is processed with bidirectional attention (as in an encoder), while the second follows causal attention (as in a decoder). This combination aims to combine contextual understanding with efficient text generation. Models such as T5, BART and UniLMv2 are part of this category.

The choice of the most suitable architecture depends directly on the nature of the task to be performed. Encoder-only models are recommended for deep textual understanding tasks, such as sentiment analysis or document classification. On the other hand, tasks that involve generating textual content - such as translation or summarization - benefit more from decoder-only or hybrid architectures. Another relevant factor is the size of the model: although larger models tend to perform better, this comes with higher computational costs (YANG *et al.*, 2024).

2.3.1 Prompt Engineering

Prompt engineering has established itself as an essential skill for the effective use of large language models (LLMs) such as ChatGPT, especially in contexts that require automation, personalization and quality control of the outputs generated. In this scenario, prompts act as instructions that guide the behavior of the models, making it possible to impose rules, define response formats, automate tasks and adapt the results to specific contexts (LIU *et al.*, 2023). Thus, prompt engineering can be understood as a new form of interactive programming, capable of shaping both the outputs and the dynamics of interaction with the LLMs (WHITE *et al.*, 2023).

As the field of prompt engineering matures, reusable structures known as prompt pattern catalogs - inspired by software engineering design patterns - have been proposed with the aim of solving recurring problems in the formulation of prompts for LLMs (WHITE *et al.*, 2023). These standards offer systematic and adaptable solutions, facilitating the transfer of knowledge

between different domains and contributing significantly to the efficient use of LLMs in complex activities such as requirements elicitation, modeling and language extension.

In addition to their technical applicability, ethical issues also stand out as essential, reinforcing the importance of developing and applying these technologies responsibly, in order to ensure positive impacts in different contexts (SAHOO *et al.*, 2024). In this sense, the use of prompt standards represents an advance in the systematization of interaction with LLMs, making their application more predictable, efficient and effective, even in specialized technical domains, such as the BPMN extension process.

However, one of the main challenges for the practical application of LLMs lies in the proper formulation of prompts - a process that is highly sensitive to linguistic nuances, requires technical knowledge and a considerable investment of time for adjustments. Small textual changes can significantly impact the quality of the results generated. This reinforces the importance of systematic approaches and tools to support prompt engineering, capable not only of optimizing model performance, but also of enabling their adoption in specific domains, such as support for modeling and extending visual languages, like BPMN (ZHOU *et al.*, 2022).

As an example of prompt engineering, we used a scenario applied to business process modeling using BPMN, as described below:

Prompt:

Given the following text describing a business process, identify and classify the involved BPMN elements, such as Task, Event, or Gateway.

Text: The customer places an order. The system checks the inventory. If available, the order is approved. Otherwise, a stock-out notice is sent.

Output:

```
Event: "Start Event"
Task: "Place order"
Task: "Check inventory"
Gateway: "Stock available?"
Task: "Approve order"
Task: "Send stock-out notice"
Event: "End Event"
```

This prompt is designed to guide a language model in identifying and classifying BPMN elements based on a textual description of a business process. By leveraging such prompts,

it is possible to automate the translation from natural language to formal process representations, streamlining the model creation and validation processes.

2.4 Chapter Summary

This chapter presents the theoretical foundations that underpin this dissertation. Initially, the concepts related to ML were discussed, with emphasis on the definition of its syntaxes and the relevance of metamodeling for the creation and evolution of languages. Next, the principles of BPM and the BPMN notation, widely adopted for business process modeling and standardized by OMG, were discussed. The formal mechanisms for extending BPMN were also examined, including its representation using XML Schema and support for creating extensions. Finally, the concepts of LLMs and Prompt Engineering were introduced, highlighting their potential as tools to support the automated and assisted creation of extensions to the BPMN language.

3 RELATED WORKS

This chapter presents the works identified in the literature that are related to the development of this dissertation, organized into sections corresponding to each stage investigated throughout the study. Section 3.1 deals with studies related to our SLR; Section 3.2 deals with work aimed at building catalogs of BPMN extensions; Section 3.3 discusses the use of LLMs in the context of supporting the extension process; and finally, Section 3.4 presents works that propose or explore processes to support the creation of BPMN extensions.

3.1 Related Works about SLR

Related work to this study involves SLR of BPMN extensions and other related modeling languages, such as UML, KAOS and iStar. The study by (BRAUN; ESSWEIN, 2014a), conducted a SLR to identify and analyze BPMN extensions, focusing on adaptations of the process modeling language that occurred between 2007 and 2014. The authors carried out a thorough analysis of the bibliographical references of the selected works, resulting in the identification of 30 relevant publications. Based on this selection, data was extracted and analyzed to assess the existing extensions and their areas of application. The study revealed that less than 20% of the extensions comply with the BPMN extension mechanism, highlighting the need for greater rigor to ensure the validity and interchangeability of the models.

In addition, (BRAUN; ESSWEIN, 2014a) emphasizes the lack of integrated methodological support for the development of extensions, especially in relation to domain analysis and comparison with standard BPMN elements. Another significant gap identified was the lack of support for resource- and data-oriented modeling aspects, suggesting the inclusion of new diagrams and views in BPMN. The study also proposes the creation of standards or guidelines for specific extension needs to improve the design process. Braun's work (BRAUN; ESSWEIN, 2014a) is particularly relevant to this research, as it takes a similar approach by running an SLR to identify and classify BPMN extensions.

(ZAROOUR *et al.*, 2020) provides a comprehensive and up-to-date analysis of BPMN extensions, covering publications from 2014 to 2018. Through an SLR, 49 BPMN extensions were identified, investigated and characterized, providing an in-depth view of practices and innovations in the field. The authors point out that although business process modeling is a central area of research due to its importance in the development of modern information systems,

BPMN, despite its wide acceptance, often needs extensions to address specific domains and improve aspects such as flexibility and security.

The study revealed that many works still do not strictly adhere to the BPMN extension mechanism, an issue previously addressed by (BRAUN; ESSWEIN, 2014a). In addition, the work highlighted the need for integrated methodological support for the development of extensions and the inclusion of new graphical elements. Both studies agree on the importance of establishing standards and guidelines for extensions, promoting greater consistency and applicability. The current research by (ZAROOUR *et al.*, 2020) not only expands but also complements Braun's findings, providing a more up-to-date and comprehensive view of BPMN extension practices and underlining the continuing need for methodological and implementation advances.

An SLR of iStar (i*) extensions (GONÇALVES *et al.*, 2018) identified 96 papers dealing with these extensions between 1995 and 2016. The papers were organized into categories and analyzed in relation to nine research questions, based on evidence collected in the selected papers. The review highlighted the different ways of representing new constructs, in line with their specific purposes. The authors emphasize the importance of ensuring consistency with existing extensions when proposing new versions, in order to avoid additional conflicts and minimize the effort required to develop iStar extensions. In addition, a catalog has been drawn up that compiles all the extensions proposed up to the last year of analysis of the review (2016) of iStar.

Another significant analysis is that of the KAOS (Knowledge Acquisition in Automated Specification) language (GONÇALVES; MONTE, 2024), which updates a previous study carried out in 2019 (SOUSA, 2019). This new investigation resulted in the identification of 50 works that represent the extensions of the language since its creation, covering the period from 1990 to 2024. The analysis was structured around seven research questions, revealing the main areas of application, the most influential authors and the evolution of proposals over time. In addition, the research evaluated both the abstract and concrete syntax of the extensions, their conservative nature, the completeness of the metamodels, the formulation of the concrete syntax and the support offered by CASE tools.

In (SILVA *et al.*, 2018b), an SLR was carried out on the UML with the aim of investigating how this language has been adapted for the creation of Domain Specific Modeling Languages (DSMLs). SLR used the snowballing technique, focusing on papers published at the International Symposium on Software Engineering for Adaptable and Self-Managing Systems

(SEAMS) between 2005 and 2007. Studies were selected based on strict inclusion and exclusion criteria, followed by data extraction and analysis to answer our research questions. In total, 16 primary studies were analyzed, addressing two main research questions. SLR reveals that the class diagram has been customized through profile-based mechanisms, providing more appropriate support for its context and promoting self-adaptability. The extension mechanisms show that this customization of the class diagram has been widely adopted.

Our study, in turn, focuses on the analysis of existing BPMN extensions over a six-year period (2019 to 2024), which had not yet been investigated by any other work, thus contributing to updating the extensions landscape. In addition, our approach integrates research questions that consider different perspectives on the approaches adopted by the studies analyzed, comparing how these approaches apply to each modeling language. This provides a more comprehensive and up-to-date view of extensions.

3.2 Related Works about BPMN Catalog

According to (CHUNG *et al.*, 2012), catalogs are a widely used solution to support software engineers in obtaining practical resources in their projects. A catalog consists of a body of knowledge, which can include general information, such as that found in books etc., or more specific knowledge, such as the BPMN developed in (CAVALCANTE *et al.*, 2025) on BPMN extensions. Often, catalogs are created with the aim of facilitating the reuse of resources in different contexts, as evidenced by (CYSNEIROS, 2007) in the elicitation of non-functional requirements.

Catalogs related to modeling language extensions have been identified, as detailed below. A related research paper was proposed in (GONÇALVES *et al.*, 2018), in which the authors present a list of extensions to the iStar modeling language, resulting from an SLR. The study aimed to identify and analyze existing extensions of the language, based on a SLR that resulted in 96 articles and 307 identified constructs. As follow-up work, the authors developed and published a catalog of iStar extensions, presented in (GONÇALVES *et al.*, 2018). This catalog, like the present work, presents a list of studies that have proposed new extensions for modeling languages, all of them obtained from a SLR.

Similarly, the work by (SOUSA, 2019) aimed to identify and analyze extensions to the KAOS language. The SLR carried out identified 35 articles in seven electronic databases, considering publications up to 2019. Subsequently, the review was updated by (GONÇALVES;

MONTE, 2024), finally reaching the identification of 50 KAOS extensions. The data collected was incorporated into the catalog of KAOS extensions proposed in (JUNIOR *et al.*, 2022), following procedures similar to those adopted in the work on iStar extensions and in this study.

The SLR on which our catalog is based is described in the work of (CAVALCANTE *et al.*, 2025). Proposed extensions to BPMN were identified in three periods, according to the papers respectively: from 2007 to 2014, from 2014 to 2018 and, more recently, from 2019 to 2025. The studies collected were incorporated into the database that makes up the catalog.

Although the aim of this work is similar to that of the aforementioned studies - to catalog modeling language extensions based on SLR - it differs in that it deals exclusively with the BPMN language, while the previous works focus on the iStar and KAOS languages. In addition, our analysis is more focused and objective, considering specific aspects such as types of extensions, compatibility between syntaxes, tool support, among others.

This process offers significant benefits, as it reduces the time and effort needed to research different sources or rely on experts. Thus, reuse becomes a more advantageous alternative than creating a new BPMN extension, as is the case in our work.

3.3 Related Works about LLMs

The increasing adoption of LLMs in different areas of knowledge has motivated the investigation of its impacts and limitations. An SLR conducted by (SALLAM, 2023) analyzed the use of ChatGPT in health-related applications, highlighting both the benefits and challenges associated with its adoption.

The results of the study indicate that 85% of the works analyzed reported benefits from using ChatGPT, highlighting among them the improvement of scientific writing, the expansion of equity and versatility in research, efficiency in tasks such as data analysis, code generation and literature reviews, as well as significant gains in professional practice such as simplification of workflows, cost savings and support for personalized medicine. These benefits also extend to the educational field, with an emphasis on personalized, problem-based learning approaches and the development of critical thinking.

Among the positive aspects discussed, the simplification of the workflow stands out, a particularly relevant aspect for this work, which proposes the use of LLMs as a support tool in the process of extension of the language BPMN. The central hypothesis is that the use guided by prompt engineering standards can promote gains in efficiency, quality and reproducibility during

the modeling and extension stages.

(KIM *et al.*, 2023) conducted a scoping review on the use of ChatGPT in academic medicine, highlighting not only the benefits but also the concerns associated with its adoption. Among the most mentioned benefits are the improvement of scientific research, the enhancement of diagnostic capabilities, and the potential transformation in academic workflow. These advances suggest that LLMs, such as ChatGPT, have the power to transform academic practices by accelerating tasks like data analysis, literature review and even simplifying complex processes, such as in personalized medicine.

However, despite the benefits, the study also points to several concerns with the use of ChatGPT and other LLMs, including ethical issues, bias, plagiarism, inaccuracy in responses, and knowledge limitations. (KIM *et al.*, 2023) argue that while ChatGPT has shown enormous potential, there are still significant challenges to overcome, especially related to security, transparency and lack of originality in the generated responses. The study suggests that as LLMs continue to be used, more precise guidelines and recommendations will be needed to ensure models are used ethically and effectively. These guidelines are fundamental, especially in sensitive areas such as medicine where integrity and accuracy are essential.

(CAIN, 2024) explores the transformative potential of Artificial Intelligence(AI) from LLMs in educational contexts, focusing on the innovative practice of prompt engineering. This practice, which is characterized by three essential components—content knowledge, critical thinking and iterative design—emerges as a key mechanism for accessing the transformative capabilities of LLMs AI in the learning process. The article traces the evolution of AI from LLMs as a tool capable of remodeling educational practices and reformulating pedagogical premises. In particular, it examines how prompt engineering practices can enhance learning by promoting personalized, engaging and inclusive educational experiences.

The authors advocate continued research on the benefits of prompt engineering practices in diverse learning contexts, while addressing potential defects, biases and ethical concerns related to the use of AI in LLMs in education. In our work, we seek to optimize this learning process by applying the engineering of prompts to improve the extension of BPMN, with the objective of making the process more efficient, ethical and adaptable to users' needs.

3.4 Related Works about Process

A significant contribution to the development of extensions in modeling languages is presented by (GONCALVES *et al.*, 2020), through the *A Process to Support iStar Extensions* (PRISE) — a systematic process designed to support the creation of iStar language extensions, widely used in Requirements Engineering projects. The PRISE appears as a response to the observation that many extensions of iStar were developed ad hoc, which caused recurring problems of incompleteness, inconsistency and semantic conflicts. To address these limitations, the process integrates consolidated guidelines from the literature, model-based development concepts, reuse of previously existing extensions and active participation of experts during creation.

The proposal was validated by recreating five existing extensions, as well as by interviews with experts and a case study involving a novice user. The results demonstrated the potential of PRISE to raise the quality and acceptability of extensions produced. One of the main differentials pointed out is the detailed modeling of the process using the BPMN notation, which facilitates its adoption by both novice users and experienced extenders. The authors also highlight that, although developed for iStar, PRISE has potential to adapt to other modeling languages, such as KAOS and UML, provided that the particularities of each language are respected.

In this context, this work seeks to continue this line of research by proposing the adaptation of PRISE for the process of extension of the BPMN language. With this, we aim to offer a systematic approach that contributes to the improvement of the quality of BPMN extensions, promoting greater consistency, reuse and alignment with the best practices of Software Engineering.

Another relevant work in the context of extensibility of modeling languages is described in (BRAUN, 2016), which proposes a systematic method for creating extensions in the Enterprise Modeling Language (EML). The approach is structured in three main stages: Domain Analysis, Extension Preparation and Extension Metamodel Design. The context analysis phase is divided into three strands: use case analysis, requirements analysis and conceptual analysis. Using the BPMN as a support language in modeling, the method aims to identify and represent the needs of stakeholders — often difficult to be explained — through textual or diagrammatic application scenarios, that make users' intentions more concrete and understandable.

The derivation of requirements can be conducted from empirical methods (such

as low fidelity interviews, observations and prototypes) or by theoretical analyses (such as systematic literature reviews or content analysis). The method also proposes a correspondence analysis, in which the derived semantic constructors are compared with the elements already supported by the language, allowing to evaluate their applicability, need for extension or discard.

In the next phase, the method adapts and extends the initial steps of the (STROPPI *et al.*, 2011) process, especially with regard to the preparation of the extension, at which time the relevant dimensions and appropriate mechanisms for the extension of language are defined. This phase culminates in the specification of the syntax (abstract and concrete) and semantics (material and formal) of the proposed extension. Finally, the extension is labeled and documented in order to facilitate its evaluation and dissemination in the community.

Like the work of (GONCALVES *et al.*, 2020), this systematic approach reinforces the importance of basing extension proposals on real needs, promoting clarity, reusability and quality. Both initiatives serve as inspiration for the proposal of this dissertation, which aims to adapt and apply these guidelines in the context of BPMN language extensions.

3.5 Chapter Summary

This chapter presented related work that is essential to support the proposal of this dissertation, organized into four main axes: SLR on extensions to BPMN, catalogs of modeling language extensions, use of LLMs to support modeling and the creation of extensions, and proposals for systematic processes. Initially, studies that carried out SLR on extensions of BPMN in three distinct periods (2007-2014, 2014-2018 and 2019-2024) were discussed, whose contributions made it possible to map the evolution of the language and identify gaps that are still present.

At the same time, catalogs based on SLR have been developed for the iStar and KAOS languages, demonstrating the value of these initiatives in promoting the reuse and organization of accumulated knowledge. This background underpins the proposal for an updated catalog of BPMN, structured with classifications by type of extension, e.g. application area and type of extension.

In addition, the chapter explored the growing use of LLMs in academic, educational and professional contexts. Recent studies highlight benefits such as efficiency in analytical tasks, code generation and support for personalized learning, while also pointing out challenges. In this scenario, prompt engineering stands out as a promising approach to ensure a more targeted,

responsible and effective use of these models.

Finally, systematic processes for creating extensions were presented, such as PRISE and Braun's method, both aimed at producing extensions based on real needs. These processes directly influence the proposal of this dissertation, which aims to adapt these approaches to the BPMN context, integrating OMG guidelines, the use of LLMs and the catalog as complementary tools.

4 AN SLR OF BPMN EXTENSIONS

This chapter presents the SLR carried out with the aim of identifying and analyzing the extensions to the BPMN language proposed between 2019 and 2024. This research complements and updates the previous review (ZAROOUR *et al.*, 2020), which covered studies up to 2018, offering a more recent and comprehensive overview of developments and trends in the extension of BPMN.

4.1 SLR Methodology

This section describes the steps taken to conduct the SLR. We followed the guidelines broadly defined by (KEELE *et al.*, 2007) and their updated version by (KITCHENHAM; BRERETON, 2013). The protocol for this SLR is detailed included defining the research questions, selecting the databases, formulating the search strings, and the inclusion and exclusion criteria for the studies. It should be noted that the protocol was documented on the Parsifal ¹ platform, ensuring the transparency and replicability of the process. The following subsections detail each of these stages.

4.1.1 Research Questions

The main research question of this SLR is: *What BPMN extensions have been proposed between 2019 and 2024?* This central question aims to identify the new BPMN extensions released in the recent period and analyze how the language has been expanded and improved. Identifying these extensions is key to providing a clear view of current trends and innovations in the field. Furthermore, it is useful for researchers in providing a systematic basis for the future development of BPMN extensions.

The research is based on identifying how many and which extensions have been introduced in the six years since the last comprehensive review on BPMN extensions, in 2018. The questions addressed were largely adapted from the studies by (ZAROOUR *et al.*, 2020) and (BRAUN; ESSWEIN, 2014a), with additions and adjustments made by the authors to meet the specific needs of this research. Our research question is refined into the following subquestions:

- **RQ1:** Is the extension proposed for a specific domain, application areas or improvement of the language itself?

¹ <https://parsif.al/>

Table 1 – Electronic Databases

Databases	URL
ACM	https://dl.acm.org/
AIIS Digital Library	https://aisel.aisnet.org/
Compendex (Engineering Village)	https://www.engineeringvillage.com/
IEEE Xplore	https://ieeexplore.ieee.org/
Science Direct	https://www.sciencedirect.com/
Scopus	https://www.scopus.com/

- **RQ1.1:** What are the specific domains targeted by the BPMN extensions?
- **RQ1.2:** What are the application areas targeted by the BPMN extensions?
- **RQ2:** Do the extensions present the meaning of the new concepts introduced? (Complete, Partial or Not Present)
- **RQ3:** The extension is presented at which syntax level (Abstract, Concrete or Both)?
- **RQ4:** Are concrete and abstract syntaxes compatible?
- **RQ5:** Are the proposed extensions conservative or non-conservative?
- **RQ6:** Which BPMN extensions have introduced new graphical representations for constructs compared to those that have adapted existing construct representations?
- **RQ7:** Which BPMN language constructs have been modified in the extensions identified?
- **RQ8:** Which tool support was proposed for BPMN extensions?
- **RQ9:** Do the proposed extensions comply with the extension mechanism specified by the OMG?

4.1.2 Database selection

An automatic search was carried out in the electronic databases listed in Table 1. These databases were selected because of their breadth and relevance, including high-quality journals and conferences in the field of software engineering.

4.1.3 Search String

The search period was defined as from 2019, since the last existing review considered up to 2018, and extends up to 2024, the year before this SLR was carried out. The search was conducted based on the terms described in Table 2. However, the search string was adapted in detail for each database searched, as shown in Table 5 (Appendix A). This adaptation took into account the specificities of each electronic search platform.

The terms used were structured into two main categories: the first includes the name of the modeling language that this study aims to investigate, along with its variations; the

Table 2 – Terms of the search

Term Classification	Related Terms
Terms	T1: "BPMN" OR "Business Process Model and Notation"
Intervention Terms	T2: "Extension" OR "Adaptation" OR "Extended" OR "Extends" OR "Extensibility" OR "Enhance" OR "Extending"

second covers terms related to extension concepts. To ensure the effectiveness of the search, a pilot test was carried out using three papers as a reference. This test successfully validated the effectiveness of the adapted search string in the selected databases. With successful validation, the returned works were exported in .bibTex format and imported into the Parsifal tool.

4.1.4 Inclusion and Exclusion Criteria

In order to guarantee the relevance and quality of the selected studies, specific inclusion and exclusion criteria were defined. These criteria aim to ensure that only relevant studies are considered to answer the research questions.

The studies to be included in the systematic review must meet the following criteria:

1. **Language:** papers written in English.
2. **Theme:** papers proposing extensions to BPMN (Business Process Model and Notation).
3. **Type of study:** Primary studies, i.e. original research that collects and analyzes data directly from the source.
4. **Publication Period:** Publications dated between 2019 and 2024.

The following criteria were established to exclude studies that did not meet the research objectives:

1. **Restricted Focus:** Studies that focus exclusively on the application of BPMN, i.e. without presenting extension proposals.
2. **Type of Study:** Secondary or tertiary studies, such as systematic reviews or systematic mappings, were not considered.
3. **Duplicity:** Duplicate papers were excluded to avoid redundancy.
4. **Type of Publication:** Works that are not academic papers have been disregarded.

4.1.5 Study selection process

The papers were selected in three stages using the Parsifal tool. In the first stage, duplicate and redundant papers were eliminated through automatic and, when necessary, manual verification, using the tool's similarity feature. The second stage involved initial analysis by

reading the titles, abstracts and keywords of the remaining papers, applying the established selection and exclusion criteria. During this phase, the papers were classified as accepted, rejected, duplicated or not classified in Parsifal. Those that were not clearly included or excluded, i.e. not classified, were kept for the final stage.

Finally, in the third stage, the remaining documents were thoroughly read and the inclusion and exclusion criteria rigorously applied. In particular, the exclusion criterion “Studies that focus exclusively on the application of BPMN, without presenting extension proposals” was applied so that papers that use BPMN without modifications to the abstract syntax (such as no changes to the metamodel or validation rules) and concrete syntax (such as existing graphical representations) were excluded, as they are not considered valid extensions of BPMN.

4.1.6 Data Extraction

Once the study selection process was complete, the data extraction stage began for the papers included in this review. At this stage, data was extracted from each of the 80 selected studies, following the criteria defined for each research question. Data extraction was facilitated by the use of the Google Spreadsheets² tool. The main objective of the data synthesis was to group the evidence collected from the selected studies, which can improve the generalization of the results.

In Google Spreadsheets, we listed all the studies in the first column, while the subsequent columns were organized according to the research questions. This organization aimed to maintain the consistency and relevance of the data in relation to the research questions. To improve the presentation of the results, we used the tool’s functionalities to generate bar charts, pie charts and other visualizations, making it easier to interpret and analyze the findings.

4.2 SLR Results

The results obtained based on our extraction from the databases will be detailed in this chapter, as shown in Figure 4.1. The key on the left-hand side of the image shows the databases followed by the number of papers identified in each one. The results generated by each step are shown by rectangles with rounded corners. The ellipses describe the steps followed during the protocol. The first data displayed is related to the identification of electronic database studies with their respective numbers.

² <https://docs.google.com/spreadsheets/create>

In this initial step, the SLR identified 1195 papers from the six selected databases, distributed as follows: 364 from the Association for Computing Machinery (ACM), 204 from the Association for Information Systems (AIS), 242 from EI Compindex, 79 from IEEE Xplore, 31 from ScienceDirect, and 261 from Scopus. However, this number was reduced to 1,181 since 14 papers were not accessible through the .bibTeX import into the Scopus database.

Step 2 is related to the removal of duplicate works, where 281 duplicate works were identified and removed, keeping 900 works. The selection of papers in steps 3 and 4 was based on the inclusion and exclusion criteria presented in the previous section. In step 3, the reviews were carried out based on the title, abstract and keywords and the papers not related to the study in question were removed, leaving 142 studies. In step 4, the text is analyzed by reading the papers in full. 15 of the papers selected in step 3 were inaccessible, resulting in 55 accepted papers. It is worth noting that, during this stage, 6 directly cited papers were identified and added to the database, bringing the total to 61 papers.

Finally, we snowballed the citations and references from the period analyzed, identifying 19 new papers, making a total of 80 papers. The data were extracted this 80 papers, whose titles and references are listed in the Appendix B in Table 6. All the data collected in this study is fully available in our Zenodo repository³. This dataset includes the document extracted from Parsifal, containing the systematized information from the review, as well as the consolidated spreadsheet with all the research questions analyzed.

Figure 4.2 illustrates the distribution of papers by year of publication. The graph shows that the years 2021 and 2022 stood out for having the highest number of extensions published. There was also a sharp drop during the pandemic period, followed by growth in the years mentioned and then a return to the normal range prior to the Coronavirus crisis.

In addition to the years of publication, the types of publications were also identified. Three categories were considered for data extraction: book chapter, conference and journal. Most of the papers publish in conferences (55%), followed by journals (42.5%) and, lastly, book chapters (2.5%).

³ <https://bit.ly/3DIwLP7>

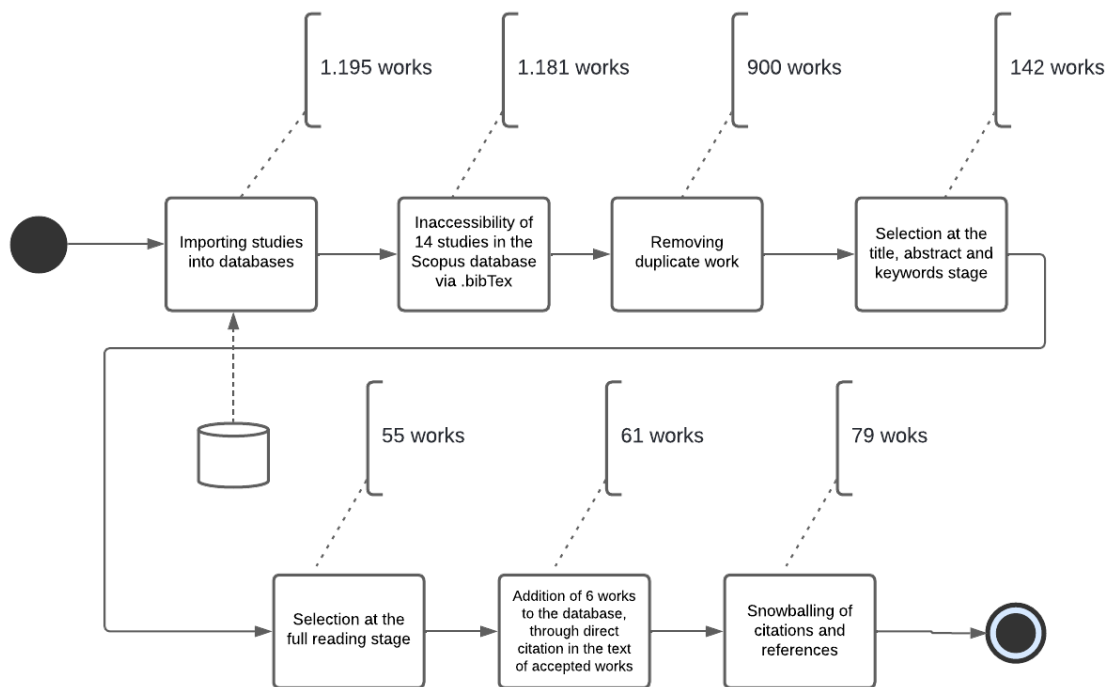


Figure 4.1 – Steps SLR Protocol

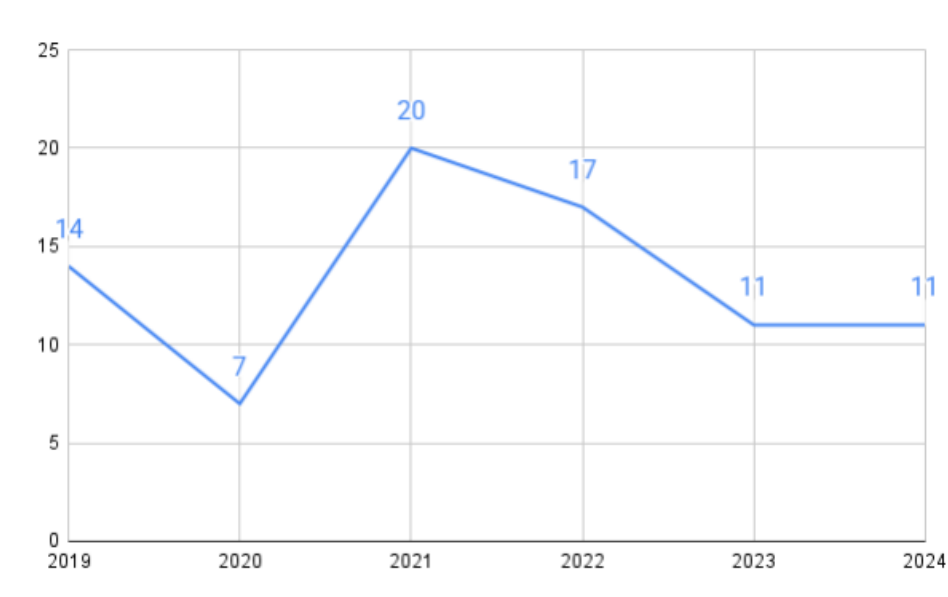


Figure 4.2 – Publication years

4.2.1 RQ1: Is the extension proposed for a specific domain, application areas or improvement of the language itself?

The information extracted to answer the second research question refers to the type of incentive that motivated the extension, whether through proposals for improvement or the search to meet a specific domain or application area. The data shown in Figure 4.3 indicates

that the majority of extensions (65%) were driven by the need to meet an application area, followed by 22.5% of extensions aimed at improving the modeling language. Finally, 12.5% of the extensions were motivated by the need to serve specific domains.

To clarify the distinction between these two concepts, it is necessary to understand that "specific domains" refer to sectors or disciplines that have clearly defined requirements, such as Healthcare, Education or Forensic Analysis. On the other hand, "application areas" cover cross-cutting themes, such as the IoT, Artificial Intelligence (AI) or Data Modeling, which can be implemented in different specific domains, adapting to the particularities of each one. For example, the AI application area can be found in specific domains such as Education, Healthcare, among others, where its techniques are adapted to meet the needs and characteristics of each field.

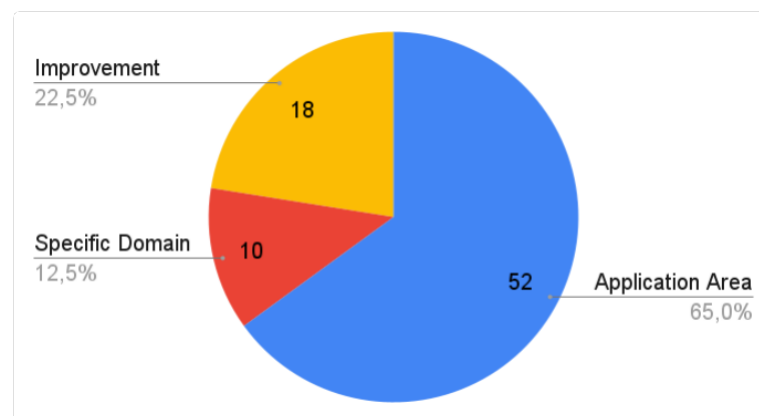


Figure 4.3 – Extensions proposal

4.2.2 RQ1.1: What are the specific domains targeted by the BPMN extensions?

Figure 4.4 presents a visual representation of the specific domains identified in 9 out of the 80 papers reviewed. These domains were classified according to the descriptions in each study, leading to a total of 4 categories. Healthcare and Forensic Analysis emerge as the most significant domains, each representing 33.3%, followed by Education at 22.2% and Electronics Manufacturing at 11.1%.

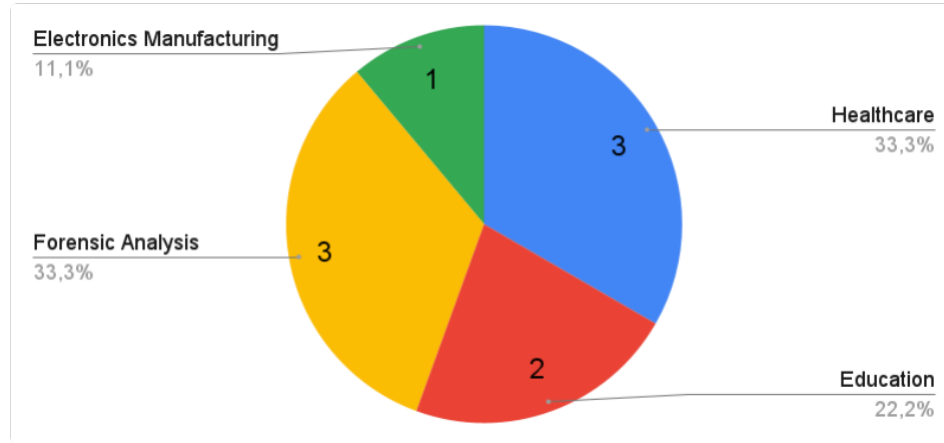


Figure 4.4 – Specific Domains targeted by BPMN extensions

4.2.3 RQ1.2: What are the application areas targeted by the BPMN extensions?

The chart presented in Figure 4.5 depicts the application areas identified across the 80 papers reviewed. These areas were classified according to the descriptions given in each paper, leading to a total of 19 unique categories. The IoT emerges as the most prominent area, representing 15% of the papers analyzed, with a total of 12 studies, highlighting its significant relevance. The second most discussed topic is Security, Distributed Systems, AI and Risk Management, with 4 studies each. This is followed by Production Engineering and Data Modeling with 3 studies each.

In addition, several areas with fewer papers were identified, including Software Architecture and Quantum Computing, each with 2 studies. Areas with only one study include: Cloud Computing, Graphical User Interfaces(GUIs), Process Management, Big Data, Continuous Processes, Business Process Governance, Physical Environment, Theory of Computing, Real-Time Business Processes and Corporate Communications.

4.2.4 RQ2: Do the extensions present the meaning of the new concepts introduced? (Complete, Partial or Not Present)

In this analysis, we sought to verify whether the proposed extensions presented clear definitions for the new extended constructs introduced. Specifically, we assessed whether the meaning of each extended construct was explicitly identified and accompanied by a detailed description, its purpose and its application in the proposed context. Figure 4.6 highlights the lack of this standardization: 55% of the works do not provide any definition, 18.8% present partial definitions, addressing only some of the new concepts, and only 26.3% offer a complete and

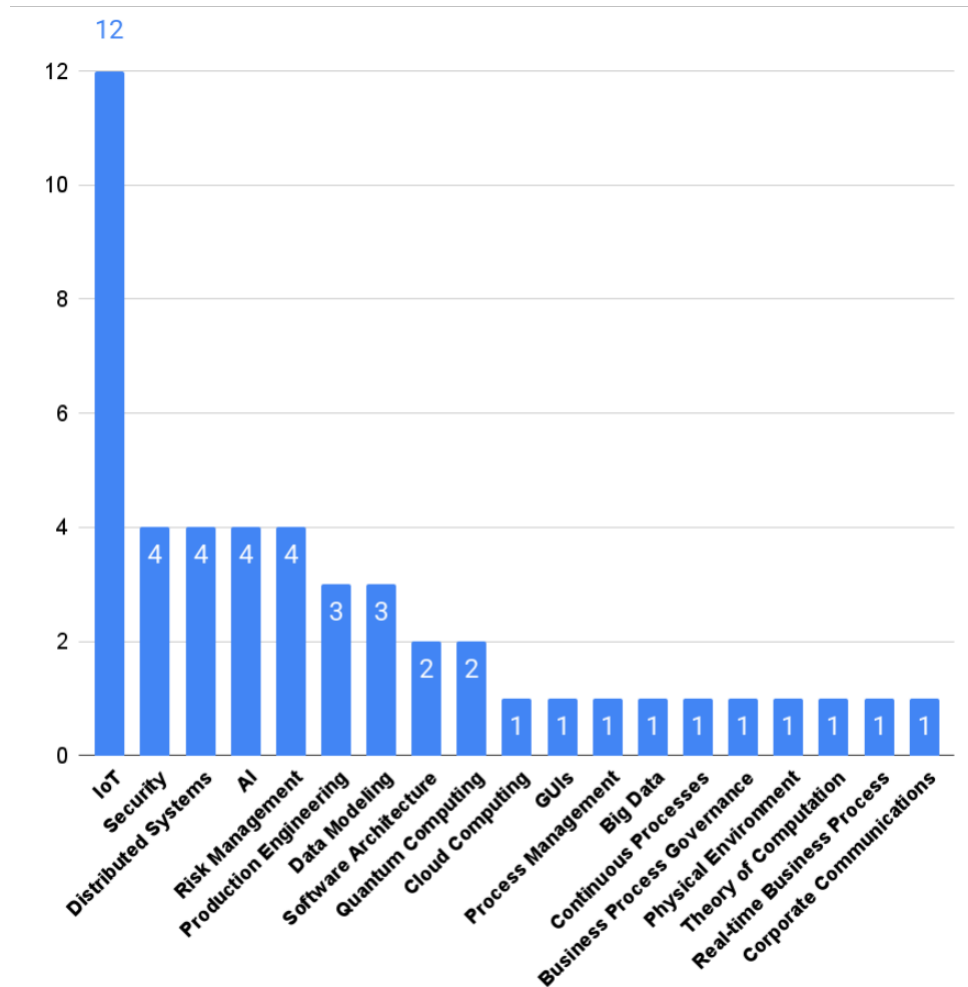


Figure 4.5 – Application areas targeted by BPMN extensions

well-defined presentation for all the constructs introduced.

4.2.5 RQ3: The extension is presented at which syntax level (Abstract, Concrete or Both)?

A survey was conducted of the types of representation used to present the extensions, classifying them into three categories:

- Abstract: When the extension was presented only through its abstract syntax, describing its structure as a metamodel.
- Concrete: when the extension has been shown exclusively visually and graphically in its new elements or modified constructs.
- Both: when the extension has been presented using both abstract and concrete representation.

This classification helps to understand how extensions have been described and presented at different levels. In this context, Figure 4.7 reveals that more than half of the papers analyzed

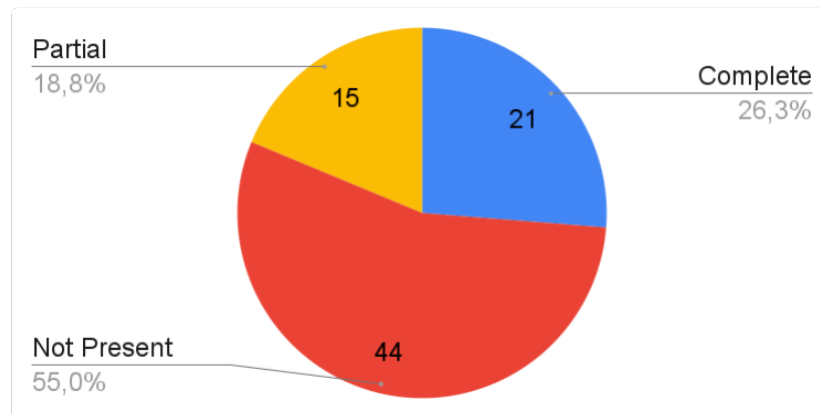


Figure 4.6 – Definition list of the new concepts introduced

(48.8%) presented both syntaxes. Next, 42,5% of the studies showed only the concrete syntax, while 8.8% showed exclusively the abstract syntax.

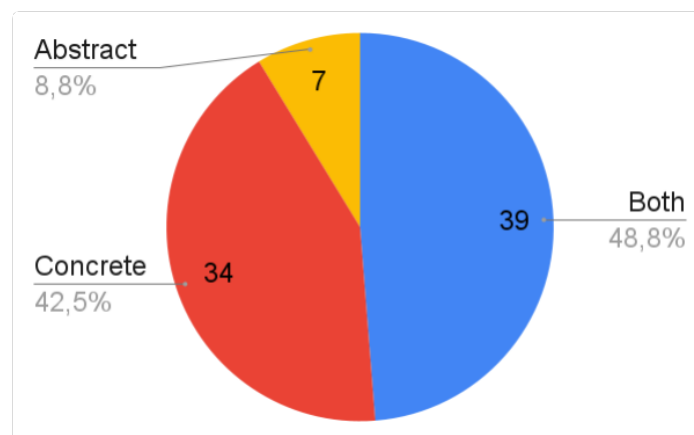


Figure 4.7 – Extensions syntax level

4.2.6 RQ4: Are concrete and abstract syntaxes compatible?

In this analysis, we extracted data from the extensions that had both syntaxes and evaluated the compatibility between the abstract and concrete syntaxes. We looked at whether the abstract syntax included all the elements and behaviors represented in the concrete syntax, and whether the concrete syntax reflected all aspects of the abstract syntax. We found that the majority of papers (76.9%) were not concerned with the correspondence between the syntaxes, while 23.1% paid attention to the compatibility between them, as shown in Figure 4.8.

We support the idea proposed by (MOODY, 2009) about the “Principle of Semiotic

Clarity”, according to which there must be a 1:1 correspondence between a graphic symbol and a specific concept. The absence of this correspondence can compromise the expressiveness and clarity of the notation, especially when relevant concepts are not adequately represented by the graphic symbols. However, we would like to emphasize that our research question focuses exclusively on the analysis of data collected from works that present both abstract and concrete elements.

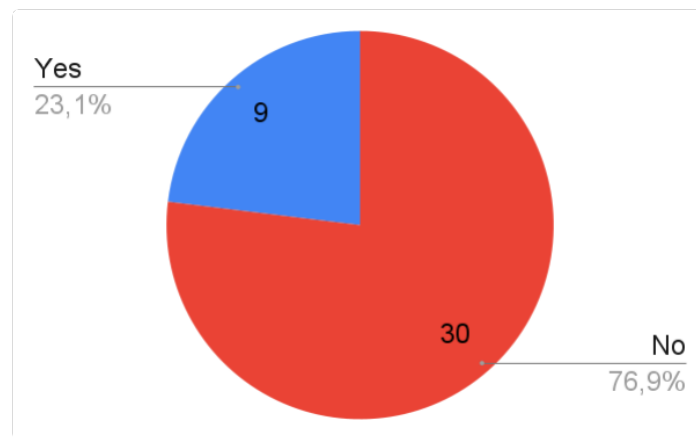


Figure 4.8 – Level of compatibility between abstract and concrete syntaxes

4.2.7 RQ5: Are the proposed extensions conservative or non-conservative?

On the topic of the level of preservation of the extension, we evaluated whether the original constructs were excluded from the construct base used, considering only the works that kept the abstract syntax. As illustrated in Figure 4.9, with the exception of two studies (4.3%), the majority of studies did not preserve the original syntax of the BPMN 2.0 version, representing 95.7% of the total.

4.2.8 RQ6: Which BPMN extensions have introduced new graphical representations for constructs compared to those that have adapted existing construct representations?

To identify how extensions approached the modification of constructs, a study was conducted that analyzed whether changes were made to existing constructs without modifying their original structure, or whether new elements were created to meet specific needs not covered by existing constructs. Figure 4.10 shows that half of the works (47.5%) chose to adapt existing

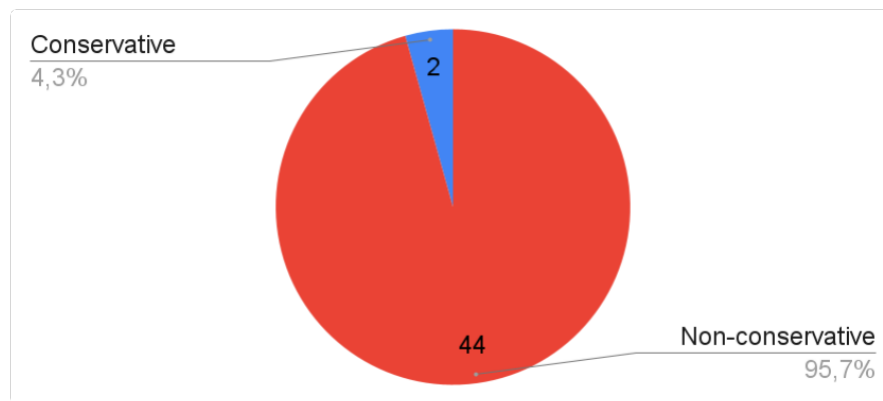


Figure 4.9 – Extension conservation level

constructs, while 15% developed new representations. Additionally, 37.5% of the extensions combined both approaches.

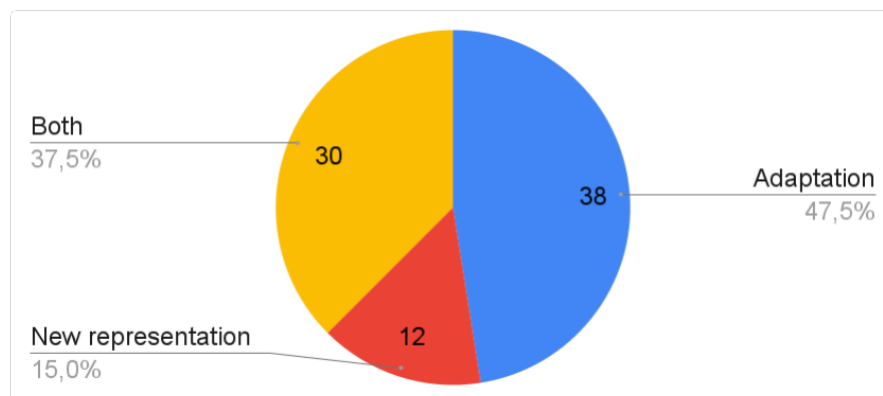


Figure 4.10 – Construct creation/adaptation level

4.2.9 RQ7: Which BPMN language constructs have been modified in the extensions identified?

Continuing the analysis of changes to constructs, it is important to highlight which elements have undergone the greatest number of adaptations. Among the most changed constructs, “Task” stands out with 43 occurrences, closely followed by “New Element”, which was introduced in 33 works. Other constructs that also underwent frequent modifications include “FlowElements” (23), “Artefact” (19), “Event” (15), “Gateway” (14), “Subprocess” (9), “Pool” (5), “Process” (2) and “Lane” (2). Finally, “Group” (1), “Text Annotation” (1), Participant (1) and Conversation (1) appear less frequently, as illustrated in Figure 4.11.

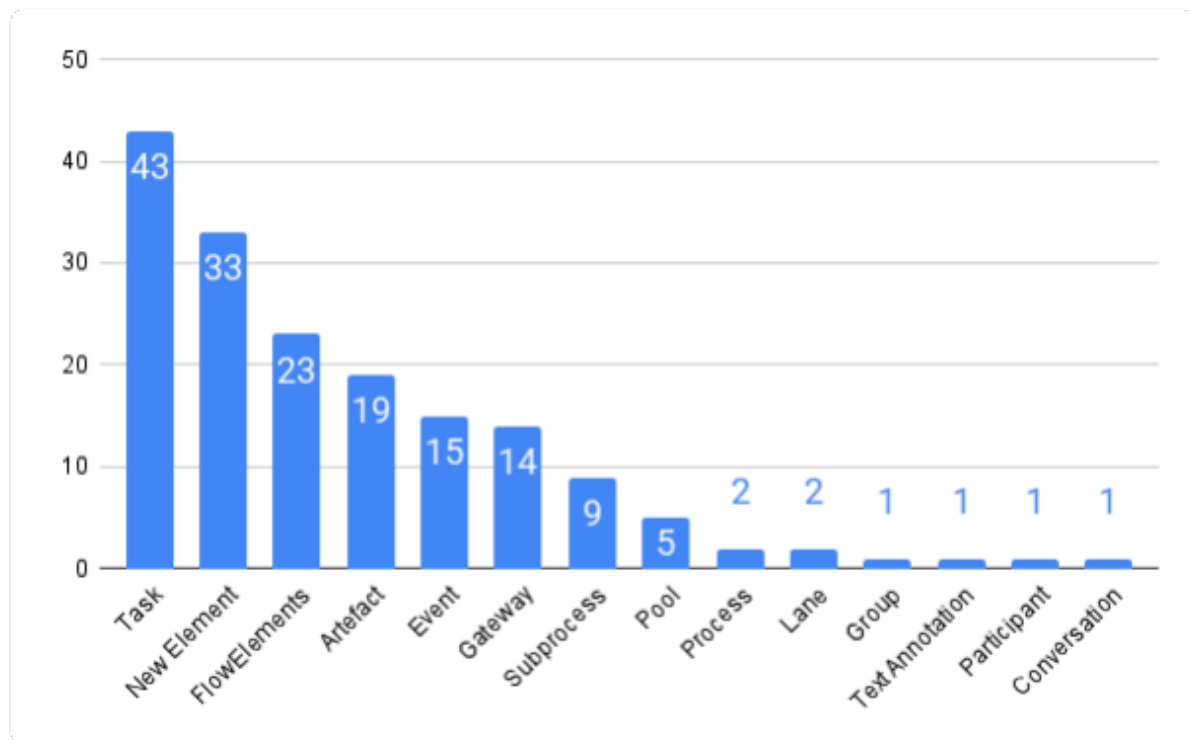


Figure 4.11 – List of constructs that have undergone modifications

4.2.10 RQ8: Which tool support was proposed for BPMN extensions?

The number of papers that reported support for the proposed extension through specific tools was investigated. The data highlights the lack of tool support, with 55% of the papers offering no to support, while only 45% provide it. This support is crucial for the adoption of the extension by the community and represents a significant added value to the work.

In Figure 4.12, it can be seen that among the authors who indicated tool support for their extensions, 11 cases proposed an original tool. In the others, existing tools or plugins were adapted to support the proposed extensions. BPMN.io Modeler was the most used tool, appearing in 9 articles. Camunda and BPMN2 Modeler were used in 6 cases each. Finally, MS Visio, Bizagi, Yaoqiang Modeler and ADOxx Modeling were used in only one extension each.

We analyzed the availability of the tools in the 11 cases in which an original tool was proposed. We found that 63.3% of the articles did not provide access links, while 36.4% provided links to their repositories on GitHub, allowing the tool to be downloaded and run locally. It should be noted that none of the original tools proposed and made available were available in a web environment.

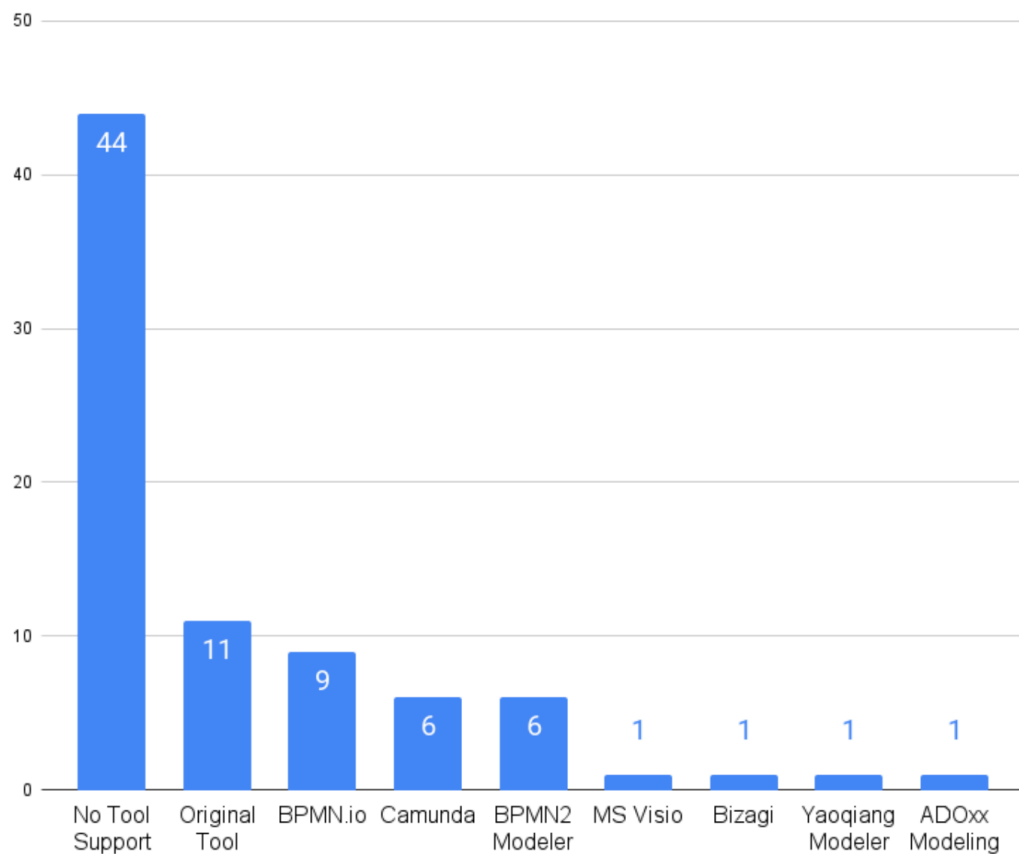


Figure 4.12 – BPMN extension if tool support

4.2.11 RQ9: Do the proposed extensions comply with the extension mechanism specified by the OMG?

The results of the survey on the compliance of papers with OMG standards reveal a worrying situation. The assessment was based on criteria established by the OMG documentation ((OMG), 2011). Among the aspects assessed is the guarantee that the basic appearance of the original elements of the BPMN language is preserved in its latest version, i.e. the original elements must remain intact. In addition, the research verified that the extension attributes do not contradict the semantics of any BPMN element, ensuring the integrity of the standard elements. Validation also included checking that the proposed extension was represented using XML Schema or XSD, as well as analyzing the correct use of the four main extension elements: `ExtensionElements`, `ExtensionDefinition`, `ExtensionAttributeDefinition`, and `ExtensionAttributeValue`.

As shown in Figure 4.13, only 3.8% of the works analyzed meet all the established criteria. In contrast, 11.3% of the works do not meet any of the necessary guidelines, while 85% meet at least one of the standards. The graph in Figure 4.14 details which works partially met the criteria, indicating the categories of guidelines that were not met. Specifically, 8.8% of the works

did not meet the XML criterion of the extension, 22.1% did not present the ExtensionElements, and 69.1% did not provide either the XML or the ExtensionElements.

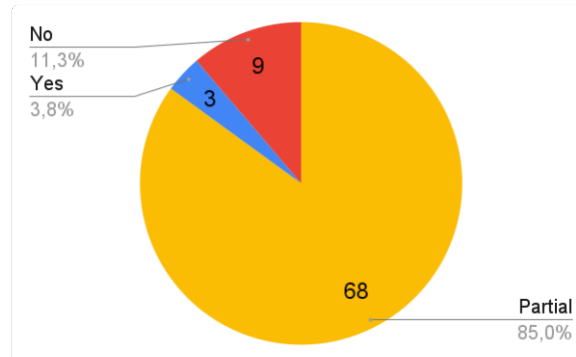


Figure 4.13 – List of works that follow OMG standards

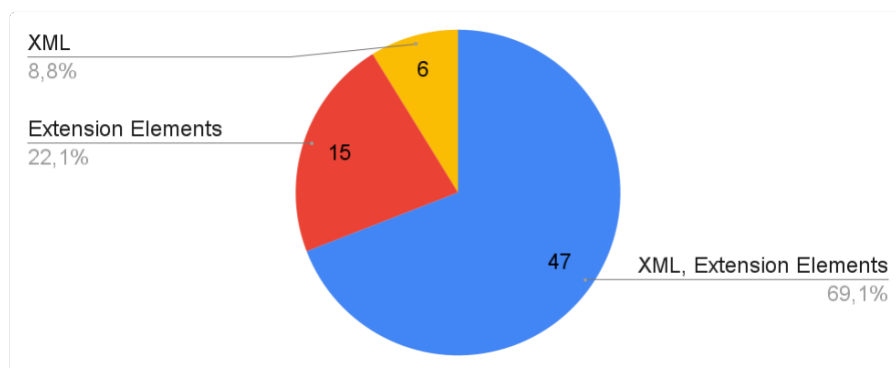


Figure 4.14 – Categories of standards not met by works that partially complied with OMG norms

4.2.12 Discussion

We presented a comprehensive analysis of BPMN extensions, identifying and correlating the most relevant approaches in the literature. An important finding is that none of the extensions examined mention the use of chatbots based on LLMs, which could interact with users during the outreach process, providing guidance on how to proceed at different stages. The adoption of such tools could be highly beneficial, as they would provide personalized assistance, speeding up the outreach process and encouraging the community to adopt and integrate these practices more effectively.

In addition, a comparative analysis with the SLR conducted by (BRAUN; ESSWEIN, 2014a) and (ZAROOUR *et al.*, 2020) shows a progressive increase in proposal new BPMN

extensions. The first SLR(BRAUN; ESSWEIN, 2014a) presented an average of approximately four publications per year over the period analyzed, while the second one (ZAROOUR *et al.*, 2020) reached an average of ten(10) publications per year. This study, on the other hand, identifies an average of thirteen(13) publications per year, showing the growing attention devoted to BPMN extensions. There is also some convergence in the areas of interest: the first SLR highlighted the focus on performance and risk management, while the second one focuses on Healthcare and IoT, where this highlights to IoT and Risk Management.

Given the large number of extensions identified, the importance of an online catalog that centralizes these extensions in an accessible repository for consultation and reuse is highlighted. Such a repository, which currently does not exist, would facilitate the identification and application of previously developed extensions, promoting reuse and contributing to the standardization and continuous evolution of BPMN. We therefore recommend the development of such a catalog, which could benefit the academic community and industry by increasing transparency and collaboration in this domain.

Among our findings, it stands out that some of the works analyzed are offshoots of previous research within the period covered by the review. Examples include the study by (OUARHIM *et al.*, 2024), which is a continuation of the work by (OUARHIM *et al.*, 2020), and that by (HASSEN *et al.*, 2019), whose extensions generated two subsequent updates: (HASSEN *et al.*, 2022a) and (HASSEN; GARGOURI, 2024). The extension of (FALCONE *et al.*, 2024) also continues the work of (DURAN *et al.*, 2022).

In addition, we analyzed the number of citations in the papers in this review to identify the relevance of these studies, as well as the levels of adoption, usefulness and applicability of the proposed extensions. The data were collected on march eleventh, two thousand and twenty five, and we selected the 15 most cited papers up to that date (WEDER *et al.*, 2020), (KALLEL *et al.*, 2021), (FALAZI *et al.*, 2019), (PULLONEN *et al.*, 2019), (ZAREEN *et al.*, 2020), (HASSEN *et al.*, 2019), (RAMOS-MERINO *et al.*, 2019), (ABOUZID; SAIDI, 2019), (EDER *et al.*, 2019), (DOMINGOS *et al.*, 2019), (NEUMANN *et al.*, 2019), (KATHREIN *et al.*, 2019), (POLANCIC, 2020), (KIRIKKAYIS *et al.*, 2023b), (KUMMER; MENDLING, 2021). The paper with the highest number of citations leads with 76(WEDER *et al.*, 2020). The average number of citations among the 15 papers analyzed is approximately 40.

During the period analyzed by this review, no studies were found that directly supported the extension of BPMN. However, it is important to highlight two studies widely cited

in the literature reviewed: those by (BRAUN; SCHLIETER, 2014) and (STROPPI *et al.*, 2011). The study by (STROPPI *et al.*, 2011) points out a significant deficiency in the BPMN extension mechanism: the lack of clear methodological guidelines for developing extensions.

To fill this gap, the authors propose a procedure based on Model-Driven Architecture (MDA), which systematizes the process of creating valid extensions to BPMN. This method is structured in four main stages: the first consists of creating a Conceptual Domain Model of the Extension (CDME) using UML. Next, a BPMN+X model is defined, which describes the extension in terms of the BPMN extension mechanism, classifying the elements of the CDME as 'BPMN concept' or 'extension concept' by means of UML stereotypes.

The third stage involves transforming the BPMN+X model into an XML Schema Extension Definition Model, which is then converted into an XML Schema Extension Definition Document. With this methodological approach, (STROPPI *et al.*, 2011) not only fills a critical gap, but also facilitates the implementation of customized extensions, broadening the applicability of BPMN in various contexts.

On the other hand, the work of (BRAUN; SCHLIETER, 2014) criticizes the approach of (STROPPI *et al.*, 2011) for not performing an adequate analysis of the semantic alignment of domain concepts with BPMN elements, as well as neglecting the analysis of requirements for extensions. The authors argue that a more detailed analysis of the conceptual phase in the development of BPMN extensions is essential for several reasons. The proposed method for developing domain-oriented BPMN extensions consists of six steps that extend the approach of (STROPPI *et al.*, 2011).

First, a requirements analysis is carried out, based on usage scenarios in the domain or literature reviews, to identify specific needs. Next, an assessment is made of whether BPMN can meet these needs, also considering other modeling languages, culminating in a decision to "approve or not" the application of BPMN. If the decision is positive, a domain ontology is built to conceptualize the context, detailing the concepts, relationships and properties.

The third stage involves checking the equivalence of the elements, analyzing whether the semantic elements correspond to the fundamental elements of BPMN, which results in a set of "extension requirements". The fourth and fifth stages, inspired by the methodology of (STROPPI *et al.*, 2011), lead to the creation of a valid BPMN extension model (abstract syntax). Finally, the last stage defines the concrete syntax of the extension, including the graphical elements in accordance with the BPMN guidelines.

Finally, we believe that improving the method proposed by (BRAUN *et al.*, 2016), which already aimed to improve the approach of (STROPPI *et al.*, 2011), could significantly contribute to expanding the number of extensions. To do this, the process would need to be structured in stages that ensure alignment with the standards set by ((OMG), 2011), preservation of the extensions in relation to the original constructs, tool support and the resolution of other gaps identified in the analysis carried out in this review.

4.3 Threats to validity of the SLR

Despite the care taken in carrying out this research, some threats to validity may influence the results and conclusions presented. This section describes concerns related to threats to validity, classified into the categories of Construct, Internal, External and Conclusion, as proposed by (WOHLIN *et al.*, 2012).

- **Construct validity:** To ensure that all relevant studies were captured, we used terms such as “BPMN”, “Business Process Model and Notation”, as well as synonyms and variations, such as “Business Process Modeling”. To cover the concept of extension, terms such as “Extension”, “Adaptability”, “Extensibility” and “Improvement” were used. However, it is possible that some relevant studies were not included due to limitations in the choice of search terms, which, when too generic, generate an excessive volume of results and make it difficult to carry out the SLR in practice. To mitigate this threat to construct validity, we carried out a pilot test of the search string, refining it based on the articles selected. In addition, we used the techniques of Forward and Backward Snowballing, which allowed us to identify additional publications on BPMN extensions in the period from 2019 to 2024, based on the references and citations of the selected studies. These approaches minimizing the risk of omitting significant work in the area.
- **Internal validity:** A concern related to the internal validity of this study lies in the possibility that subjective decisions influenced the process of article selection, quality analysis and data extraction. This subjectivity could hinder the objective application of inclusion/exclusion criteria and unbiased data extraction. To minimize errors in selection, quality analysis and data extraction, the steps of this SLR were carried out using a peer review approach. If there were any conflicts, all the authors discussed and resolved the differences, seeking to mitigate the threats posed by personal bias in the interpretation of the studies.

- **External validity:** One of the concerns regarding the external validity of this study relates to the generalizability of the results of the SLR, which is related to the degree to which the primary studies are representative for the topic under analysis. To mitigate external threats, the definition of the search was carried out after several testing searches, and was validated by the consensus of all the authors. This approach ensured greater comprehensiveness and rigor in the selection of relevant articles.
- **Conclusion Validity:** The methodology used, as presented in (KITCHENHAM; BRERETON, 2013), recognizes that not all relevant primary studies can be identified. There is a possibility that some studies that were excluded in this review could have been included. To mitigate this threat, the selection process and the inclusion and exclusion criteria were carefully designed and discussed among the authors, with the aim of minimizing the risk of excluding relevant studies.

5 A CATALOG OF BPMN EXTENSIONS

This chapter discusses the development of an online catalog of the BPMN extensions identified in the SLR of the previous chapter. The catalog aims to organize, make available and facilitate access to the extensions, offering a practical and up-to-date tool that complements previous studies and supports researchers and professionals in consulting and applying these extensions.

5.1 BPMN Catalog Methodology

This section describes the steps taken to develop the BPMN language extensions catalog. The entire process is presented in detail, from the definition of the objectives and research questions, through the extraction of data relating to the extensions, implementation and feeding of the catalog, to its subsequent validation. The following subsections explain each of these stages in an individual and structured way.

5.1.1 *Objectives*

5.1.1.1 *General Objective*

To develop and evaluate an online catalog of extensions for the BPMN modeling language, with the aim of providing an organized, accessible, and useful tool for consulting and registering extensions, contributing to the BPMN community.

5.1.1.2 *Specific Objectives*

- Propose an online tool for cataloging and consulting BPMN extensions.
- Evaluate the accuracy and effectiveness of the catalog.

5.1.2 *Research Questions*

The research questions of this study address the central aspects related to the catalog of BPMN language extensions.

- **RQ1: How can the existing BPMN language extensions be catalogued?**

This question investigates the main extensions proposed for the BPMN language, with

a view to identifying and organizing them in a catalog. In addition, essential criteria are analyzed to ensure that the most relevant information is included in the catalog in a structured way.

– **RQ2: How efficient is the catalog developed for BPMN extensions?**

This question assesses the efficiency of the catalog developed, considering its ability to facilitate the organization and consultation of BPMN extensions, promoting a more effective and intuitive interaction for users. To do this, metrics were used based on the UTAUT (Unified Theory of Acceptance and Use of Technology) model, which includes factors such as intention to use, performance expectancy, effort expectancy, and facilitating conditions.

5.1.3 Study Steps

This section describes the steps followed to develop the catalog of BPMN extensions. The procedures are presented below, highlighting that the results of the SLR on BPMN extensions (CAVALCANTE *et al.*, 2025) served as the starting point for the activities carried out in this work.

5.1.3.1 Extracting the Extensions

The extraction of existing extensions is based on a database drawn up from the SLR research questions (CAVALCANTE *et al.*, 2025), containing all the information needed to complete the catalog. In addition, this stage includes storing the data of all the builders, which will now be preserved to be attached to their respective publication — an aspect that had not been addressed in the review.

5.1.3.2 Catalog Implementation

The BPMN extensions catalog is developed as a web version, using the PHP¹ language with the Laravel² framework. The database used is MySQL³, and the application is containerized with Docker⁴. The catalog is structured in advance to allow the inclusion of data using forms, enabling the registration of the fields needed to feed each publication.

¹ <https://www.php.net/>

² <https://laravel.com/>

³ <https://www.mysql.com/>

⁴ <https://www.docker.com/>

These fields are based on the research questions addressed by the SLR (CAVALCANTE *et al.*, 2025), providing better organization and availability of information. In addition, the catalog differentiates itself by including a review mechanism that lists all the extended constructors, allowing them to be reused by the community.

5.1.3.3 *Inclusion in the Catalog*

With the information gathered and the catalog available, the final update is made to include the necessary data, completing the process of registering extensions. Each publication receives the information according to the topics listed below, plus the addition of image-type attachments representing the extended constructors in cases where there is concrete extended syntax, accompanied by an example of their use.

5.1.3.4 *Catalog Evaluation*

To assess the acceptance of the BPMN Extensions Catalog, a survey (KITCHENHAM *et al.*, 2002) based on the UTAUT model will be conducted. The research involved 13 master's students, who answered a structured questionnaire based on the model's main constructs: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions. The answers were recorded on a five-point Likert scale, ranging from "Strongly Disagree" to "Strongly Agree". Before the final application, a pilot test was carried out with five master's students, whose suggestions were incorporated to ensure the clarity and suitability of the questionnaire.

The suggestions obtained were analyzed and incorporated into the final version of the questionnaire, ensuring its clarity and suitability. The UTAUT model, proposed by (VENKATESH *et al.*, 2003), is widely used to study technology adoption and integrates eight theories on technological acceptance, including the Theory of Reasoned Action (TRA), the Technology Acceptance Model (TAM), and the Theory of Planned Behavior (TPB). Its main objective is to understand the factors that influence the intention to use a system, with four constructs having a direct impact on this process: Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions.

Thus, this study seeks to identify whether these factors influence the adoption of the catalog, helping to understand its acceptance and feasibility of implementation in the BPMN community.

5.2 BPMN Catalog Results

5.2.1 Presentation of the BPMN extensions catalog

The catalog, available online⁵. In the catalog, you can view the complete list of registered extensions and select any of them to access details with all the registered information and their respective builders. The tool offers search capabilities by author or title and allows filters to be applied by various categories, such as application area, specific domain, proposed improvements, presentation of new constructors, type of conservativity, compliance with OMG standards, type of representation, availability of support tools, syntax level and compatibility between different syntaxes.

Figure 5.1 shows the list of extensions cataloged, including the information mentioned and a link to access the full details of each extension. Figure 5.2 shows the details of an extension, where the title is clickable and provides all the information previously described. Figure 5.3 highlights the extended constructs related to the extension.

BPMN Catalog Publications Constructs Suggest Papers Team Login ▾				
Publications List				
Search by Author, Title or Filter by Categories				
Author or Title Search		and/or	Filter By ▾	Search
ID	Title	Authors	Year	Source
2	A BPMN Extension for Business Process Outsourcing to the Cloud	Zarour, Karim and Benmerzoug, Djamel and Guermouche, Nawal and Drira, Khalil	2019	New Knowledge in Information Systems and Technologies (WorldCIST 19 2019)
3	A BPMN extension to model Inter-organizational processes in Industry 4.0	Vitor Ribeiro, João Barata, Paulo Rupino da Cunha	2021	Sensors
1	A BPMN-based language for modeling corporate communications	Gregor Polančič, Boštjan Orban	2019	Computer Standards & Interfaces
4	A Business Process and Data Modelling Approach to Enhance Cyber Security in Smart Cities	Josef Horalek, Tereza Otcenaskova, Vladimir Sobeslav, Petr Tucnik	2023	International Journal of Advanced Computer Science and Applications (IJACSA)
5	A Business Process Modeling Notation Extension for Real Time Handling-Application to Novel Coronavirus (2019-nCoV) management process	Ouarhim, Asma and Lakhrouit, Jihane and Baina, Karim	2020	Journal of Systems and Software
6	A Family of Experiments to generate Graphical User Interfaces from BPMN Models with Stereotypes	Eduardo Díaz, José Ignacio Panach, Silvia Rueda, Damiano Distanto	2021	Software and Systems Modelling
7	A Lightweight BPMN Extension for Business Process-Oriented Requirements Engineering	Intrigila, Benedetto and Della Penna, Giuseppe and D'ambrogio, Andrea	2021	Journal of Biomedical Informatics
8	A Migration Framework for Active BPMN Processes in Healthcare	Matteo Mantovani, Emanuele Chini, Carlo Combi	2024	IEEE Access
9	A Model Based Framework for IoT-Aware Business Process Management	Bocciarelli, Paolo and D'Ambrogio, Andrea and Panetti, Tommaso	2023	International Journal of Information Management
10	A Multi-criteria Evaluation Approach for Selecting a Sensitive Business Process Modeling Language for Knowledge Management	Ben Hassen, Mariam and Turki, Mohamed and Gargouri, Faleh	2019	International Journal of Information Management

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Figure 5.1 – BPMN Extensions Listing Page

⁵ <http://bpmnextensionscatalogue.quixada.ufc.br/>

BPMN Catalog Publications Constructs Suggest Papers Team Login ▾

A BPMN Extension for Business Process Outsourcing to the Cloud

Author: Zarour, Karim and Benmerzoug, Djamel and Guermouche, Nawal and Drira, Khalil
 Type: Conference
 Name: New Knowledge in Information Systems and Technologies
 Year: 2019
 Description:

Category Application Area	Concepts Not Present	Conservative Non-Conservative	Constructors Modified New element
OMG Yes	Representation New Representation	Specific Domain/Application Area Cloud Computing	Support Yes
Syntaxes Compatible No	Tool MS Visio and Bizagi		Syntax Level Both

Figure 5.2 – BPMN Extension Details Page – Information

Constructs				
ID	Concept	Type	Image	Option
1	Security	entity		Detail
2	Compliance	entity		Detail
3	Cost	entity		Detail
4	Performance	entity		Detail

[Suggests Update Info](#)
Status: Approved

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Figure 5.3 – BPMN Extension Details Page – Constructs

In addition, more detailed information can be accessed via the “Details” option, as illustrated in Figure 5.4. This detailed information offers a comprehensive view of each extension and its specific features, including:

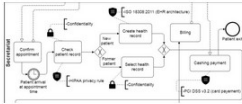
- **Description:** Explains the concept of the constructor as presented in the source work.
- **Shape:** Describes the visual representation and structure of the constructor.

BPMN Catalog Publications Constructs Suggest Papers Team Login ▾

Construct: 1 - Security

Related Extension: 01 - A BPMN Extension for Business Process Outsourcing to the Cloud (Zarour, Karim and Benmerzoug, Djamel and Guermouche, Nawal and Drira, Khalil)

Priorization: 1

Description An object that handles Security as an activity type Form Icons Classification New Representation New Representation	Notation:  Conflicts No conflicts
Example: 	

This catalog manager is open-sourced software licensed under the [MIT License](#). Copyright (c) Tiago Heineck, Milene Cavalcante, Enyo Gonçalves, João Araújo

Figure 5.4 – Construct Details Page

BPMN Catalog Publications Constructs Suggest Papers Team Login ▾

Construct List

Search by Concept -- Filter By Area -- -- Filter By Form -- -- Filter By Type -- Search Clear

Concept	Form	Type	Image	Priorization
Performance Icon	Icons	entity		1
Production Task	Industry 4.0 Elements	entity		1
Logistics Task	Industry 4.0 Elements	entity		1
Traceable Task	Industry 4.0 Elements	entity		1
Private Task	Industry 4.0 Elements	entity		1
Touchpoint Task	Industry 4.0 Elements	entity		1

Figure 5.5 – Extended Constructs List Page

- **Classification:** Indicates whether the constructor represents an adaptation of an existing element or a novel representation.
- **Notation:** Presents the graphical notation or symbol used for the constructor.
- **Example:** Provides a practical example demonstrating the use of the constructor in a BPMN process.
- **Conflicts:** Identifies any potential incompatibilities or overlaps with other BPMN constructors.

Similar to the list of extensions, you can also access the list of all cataloged con-

structors. The tool allows you to search for constructors by name or apply specific filters, such as Shape (constructor format), Type (Node or Link) and Classification (Adaptation or New Representation), as shown in Figure 5.5. By clicking on the name of a constructor, detailed information is displayed on a dedicated page, the same one accessed via publications, as shown in Figure 5.4.

5.2.2 Evaluation of the BPMN Catalog

The survey was applied after an introductory presentation on BPMN, its extensions and the usual process of searching for these extensions in an ad-hoc manner, followed by a practical demonstration using the catalog developed. Seventeen Master's students in Computer Science took part, interacting with simulated scenarios designed to provide an initial experience of using BPMN extensions. Prior to this, a pilot test was conducted with five other master's students in order to adjust the collection instrument. The questionnaire was structured based on the UTAUT model, covering aspects such as prior knowledge of BPMN and extensions, intention to use, performance expectations, effort expectations, facilitating conditions and use.

Of the participants, 82.4% (14) said they had beginner-level knowledge of BPMN, while 11.8% (2) were intermediate and only 5.9% (1) said they had advanced knowledge. With regard to knowledge of BPMN extensions, 88.2% (15) classified themselves as beginners and 11.8% (2) as intermediate, with no participants having advanced knowledge in this category.

The survey data was analyzed based on the levels of the 5-point Likert scale used in the questionnaire, in which participants indicated their degree of agreement with each statement (1 = Strongly disagree; 5 = Strongly agree). The answers were quantified in order to calculate averages per construct of the UTAUT model, allowing a numerical assessment of users' perceptions of the catalog. However, some statements were deliberately worded negatively, such as: "I don't want to use the catalog" or "The catalog will not be compatible with other tools I use". In these cases, the answers were reversed (5 = Strongly disagree; 1 = Strongly agree) in order to maintain the interpretative consistency of the scores - i.e. to ensure that higher values represent more positive perceptions across all items.

After this normalization, the averages per participant and per construct were calculated, covering: intention to use, expectation of performance, expectation of effort, facilitating conditions and use. Based on this normalization, the interpretation of the results considers that the closer the averages obtained are to 5, the more positive the participants' perceptions of the

catalog are.

Tables 3 and 4 provide a comprehensive overview of the participants' perceptions, based on the answers collected through the survey. Table 3 shows the detailed results by item, enabling a more precise analysis of individual opinions on specific aspects evaluated in the UTAUT model. In turn, Table 4 consolidates this information by construct, facilitating a general understanding of the levels of acceptance and perceived usability. This data helps to assess the potential for adoption of the catalog and points to areas that require improvement, such as strengthening support conditions and adopting strategies to encourage continued use of the tool.

The results obtained show a positive intention to use the catalog on the part of the participants, with an overall average of 4.44 in the Intention to Use construct, indicating agreement with statements that recognize the relevance and usefulness of the tool. Items such as "The catalog is relevant for searching for BPMN extensions" (average of 4.82) and "The catalog will make the search process more agile" (4.59) reinforce this perception. In addition, Performance Expectation had an average of 4.50, showing that users believe the catalog can improve their productivity, speed up tasks and increase the chances of finding suitable extensions, as indicated by the high value of the item "Using the catalog will give me a better chance of finding the extension I need" (4.88).

The perception of ease of use was also favorable, with an average of 4.18 in Effort Expectation, suggesting that participants find the catalog accessible and intuitive. However, the Facilitating Conditions construct obtained a lower average (3.51), with a negative highlight for the item "A specific person or group will be available to help me when I have difficulties" (2.88), pointing to the need for institutional or technical support.

Finally, although the intention to use was high, the average for the Use construct was 3.26, indicating a more moderate willingness to adopt the catalog immediately. This difference may be related to factors pointed out in the statements themselves, such as uncertainty about using it in the coming months (averages of 2.65 and 2.59), the relatively low level of intention to exploit its functionalities to the full (2.94), or even hesitation in adopting it as a first choice over other options (3.76). These results suggest possible external barriers, such as a lack of ongoing familiarity with the tool, the absence of institutional incentives or not perceiving an immediate need to use it at the moment.

Table 3 – Descriptive Statistics of UTAUT Constructs and Associated Items

Construct [Item]	Average	Standard Deviation
Intention to use [I don't want to use the catalog]	3.65	1.00
Intention to use [Although it can be useful, using the catalog can make work more complex]	4.41	0.87
Intention to use [Using the catalog will not make extension development more productive]	4.71	0.47
Intention to use [The catalog is relevant to the BPMN extension search]	4.82	0.39
Intention to use [The catalog will make the extension search process more agile]	4.59	0.71
Performance expectation [I believe the catalog will be useful when searching for BPMN extensions]	4.71	0.47
Performance expectation [Using the catalog will allow me to perform activities more quickly]	4.18	1.07
Performance expectation [Using the catalog will increase my productivity]	4.24	1.09
Performance expectation [Using the catalog will give me a better chance of finding the extension I need]	4.88	0.49
Effort expectation [My interaction with the catalog will be clear and easy to understand]	4.12	0.60
Effort expectation [It will be easy for me to become a catalog user]	3.71	0.99
Effort expectation [I think the catalog will be easy to use]	4.53	0.62
Effort expectation [Learning to use the catalog will be easy for me]	4.35	0.70
Facilitating conditions [I will have the necessary resources to use the catalog]	4.29	0.77
Facilitating conditions [I will have the necessary knowledge to use the catalog]	3.65	0.93
Facilitating conditions [The catalog will not be compatible with other tools I use]	3.24	1.09
Facilitating conditions [A specific person or group will be available to help me when I have difficulties]	2.88	0.70
Use [I plan to use the catalog in the coming months]	2.65	1.00
Use [I will use the catalog in the coming months]	2.59	0.87
Use [I will explore the catalog to the fullest]	2.94	1.03
Use [I plan to discover new ways to use the catalog when looking for BPMN extensions]	3.35	1.00
Use [Even when other options are available, the catalog will always be my first choice]	3.76	0.90
Use [I plan to make the most of the information provided by the catalog about BPMN extensions]	3.88	0.93
Use [I plan to integrate the catalog when developing BPMN extensions]	3.65	0.86

Table 4 – Descriptive Statistics of UTAUT Constructs

Construct	Average (Mean)	Standard Deviation
Intention to Use	4.44	0.69
Performance Expectation	4.50	0.78
Effort Expectation	4.18	0.73
Facilitating Conditions	3.51	0.87
Use	3.26	0.93

5.3 Threats to Validity

This section discusses the main threats to the validity of the survey instrument used in this study, based on the four criteria proposed by (KITCHENHAM; PFLEEGER, 2002): face validity, content validity, conclusion validity and construct validity.

Face Validity refers to the preliminary assessment of the clarity, readability and comprehensibility of the instrument's items by people with no prior knowledge of the subject. For this stage, we applied the pilot of the questionnaire described in the previous section, collecting the results which led to improvements in the fluidity and wording of the questions.

Content Quality refers to the qualitative assessment of the adequacy, comprehensiveness and relevance of the items by experts or people familiar with the topic. To this end, the form was reviewed by two experts in the field, ensuring that all the important dimensions of the object of study were covered.

Completion Validity refers to the instrument's ability to achieve its objective, considering the reliability of the data collected and the possibility of carrying out robust statistical analyses. In this study, 17 students took part in the face-to-face application of the questionnaire. Although this number was sufficient to obtain an initial overview of usability, it limits the ability to make more in-depth statistical inferences, thus restricting the validity of the quantitative conclusions.

Construct Validity assesses whether the instrument effectively measures the constructs it sets out to investigate in the context of the research. To mitigate threats related to this validity, the questionnaire was developed with clear, objective questions and was reviewed and validated in the pilot stages, ensuring that the answers consistently reflected the participants' perceptions of the catalog's usability.

Despite the care taken, these threats to validity should be considered when interpreting the results, and indicate the need for future research with larger samples and complementary methodologies to strengthen the evidence obtained.

6 METHODOLOGY

This chapter presents the methodology proposed to investigate the use of LLMs to support the creation of extensions to the BPMN language. At the same time, the development of a systematic extension process is described, based on the adaptation of the methods PRISE (GONCALVES *et al.*, 2020) and Braun (BRAUN, 2016) methods, in accordance with the technical guidelines established by the OMG. The stages of this methodology are summarized in Figure 6.1.

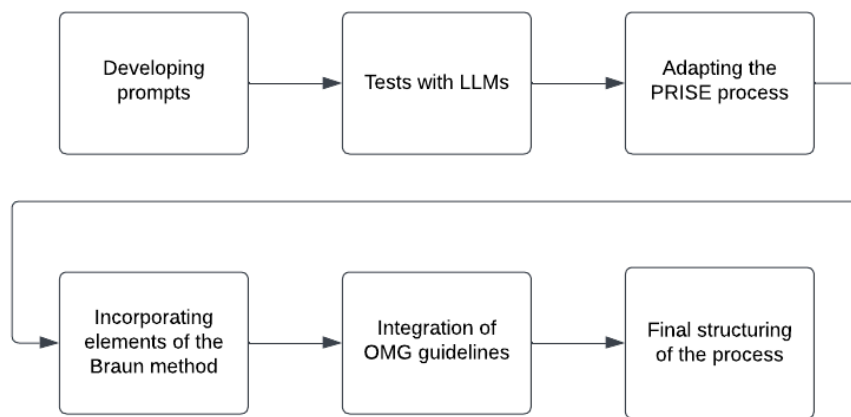


Figure 6.1 – Steps in the methodology

6.1 Methodology for using LLMs in the proposed extension of BPMN

This section describes the methodological procedures adopted for the development and evaluation of an assistive approach based on LLMs, aimed at supporting the creation of extensions to the BPMN language. The methodology is organized into the following stages:

6.1.1 Elaboration of Structured Prompts

Structured prompts are developed based on prompt engineering techniques, organized in progressive steps to guide LLMs in the creation and validation of BPMN extension proposals. The main elements considered in the prompts include:

- **Contextualization of the extension task:** Presentation of the scenario, motivation and need to adapt the BPMN notation to the user’s specific context;
- **Specification of OMG requirements:** Directing the model to the formal guidelines of the

OMG, including the structure of the metamodel and the valid extension criteria;

- **Extension XML structure generation:** Explicit request for the model to produce the XML code corresponding to the extension proposal, compatible with the OMG schemas;
- **Expert review simulation:** Inclusion of an additional prompt in which LLMs are instructed to assume the role of a process modeling expert in order to critically review the generated proposal, point out inconsistencies and suggest improvements.

6.1.2 *Executing Interactions with LLMs*

The prompts are applied in isolated sessions, always starting from scratch, so as to avoid any influence from the accumulated context of previous interactions. At this stage, the focus is exclusively on formulating and applying the prompts, without first considering which language model will be used. The identification of the models used is only considered in the subsequent evaluation stage, allowing for an impartial analysis based exclusively on the results produced. For each proposed scenario:

- The complete answers generated are recorded in full;
- Multiple versions of the answers are obtained (when necessary), in order to observe variations and consistency.

6.1.3 *Analysis of Generated Responses*

The outputs generated by LLMs, based on the structured prompts, are analyzed according to the elements requested and their quality in relation to the defined criteria. The analysis takes into account:

- **Compliance with OMG guidelines:** Evaluation of the alignment of the proposals with the OMG principles, especially with regard to the structure of the abstract syntax and the appropriate use of the BPMN metamodel;
- **Validation of the generated XML structure:** Verification of the presence, formatting and correctness of the resulting XML code, based on the structure required by the OMG schemas;
- **Completeness of proposal elements:** Checking the presence and coherence of the expected components - motivation for the extension, definition of the new construct, proposal of abstract and concrete syntax, as well as a contextualized example of use;
- **Clarity and usability of content:** Analysis of the language used, comprehensibility of

the explanations and usefulness of the answers for users with different profiles, including non-experts;

- **Quality of the expert review simulation:** Critical analysis of the answers provided by the model in the role of expert reviewer, checking whether relevant inconsistencies have been identified and whether suggestions for refinement are pertinent and constructive.

6.1.4 *Refinement of Prompts*

Based on the analysis of the results and the simulated feedback, the prompts are adjusted to:

- Reduce ambiguity;
- Provide more specific instructions to the model;
- Increase the accuracy and conformity of the answers generated.

6.1.5 *Evaluation of the Optimized Prompt*

After selecting the optimized prompt, it was applied to an unprecedented set of test scenarios, different from those used during the adjustment and refinement phases. The evaluation was based on the responses generated by three different language models: DeepSeek¹, Perplexity.ai² and Copilot³.

To measure the effectiveness of the optimized prompt comprehensively, a quantitative analysis was conducted using several standard performance metrics, including:

- **Precision:** proportion of correct answers in relation to the total number of answers generated, indicating how often the model gets its predictions or classifications right;
- **Recall:** the model's ability to correctly identify all relevant instances, i.e. the proportion of true positives detected in relation to the total number of relevant cases;
- **F1 Score:** harmonic mean between precision and recall, offering a single metric that balances both dimensions - the higher, the better the overall performance;
- **Response time:** average time required for the model to generate a response to a given input, an important factor in assessing efficiency and practical applicability.

In addition to quantitative metrics, qualitative aspects were evaluated, such as the **stability** and **consistency** of the responses. To do this, each model was subjected to multiple

¹ <https://www.deepseek.com/>

² <https://www.perplexity.ai/>

³ <https://copilot.microsoft.com>

executions with the optimized prompt, always in isolated sessions, in order to avoid interference caused by interaction history.

The results were compared between the three models, allowing us to see how each LLMs behaves in the same scenarios and under the same input conditions. This comparison aims to identify differences in the overall quality of the responses, as well as the generalization capacity and robustness of the optimized prompt in different contexts and platforms.

6.1.6 Documentation and Reproducibility

The entire process is documented to ensure reproducibility, including:

- Record of scenarios used;
- Versions of prompts and their outputs;
- Classification of errors observed and adjustments made.

6.2 Methodology for building the BPMN extension process

This section describes the method proposed to support the process of extending the BPMN modeling language. The method is based on consolidated language extension approaches, including the PRISE (GONCALVES *et al.*, 2020) process, aimed at extending the iStar language, and Braun’s (BRAUN, 2016) method, used to create extensions for Enterprise Modeling Languages (EMLs). In addition, the proposed process incorporates OMG guidelines and contemporary tools, such as LLMs and an online catalog of BPMN extensions.

6.2.1 Study and Adaptation of the PRISE Process

The process PRISE (GONCALVES *et al.*, 2020) process, originally developed for the iStar language extension, serves as the basis for identifying steps and practices applicable to the BPMN context. The analysis considers:

- The methodological structure and the sequence of its steps;
- The artifacts generated (such as conceptual descriptions and validations);
- The role of experts and the reuse of concepts in the process.

Based on this analysis, elements compatible with the objectives of the proposal are selected and adapted, respecting the particularities of the BPMN language and its possibilities for extension.

6.2.2 Analysis of the Braun Method

The Braun (BRAUN, 2016) method, aimed at creating extensions for MLEs, complements the construction process. The study of the method focuses on:

- The domain analysis phase;
- The technical preparation of the extension with a focus on the metamodel;
- The syntactic and semantic validation mechanisms.

The elements identified are combined with the process derived from PRISE, with emphasis on conceptual formalization and technical alignment with the BPMN metamodels.

6.2.3 Incorporation of the OMG Guidelines

The guidelines established by OMG for extensions to BPMN are gathered and organized as formal requirements for the process. These guidelines provide guidance:

- The structuring of technical artifacts (such as valid XML schemas);
- Maintaining compatibility with the official metamodel;
- Presentation of the extension's metamodel.

These guidelines are considered both in the generation of artifacts and in the technical validation of extensions.

6.2.4 Planning the Integration of LLMs and Catalog of Extensions

As part of the structuring of the proposed method, the integration of support resources that can contribute to the effectiveness and practicality of the BPMN extension process is planned. These resources include the LLMs and an online catalog of extensions to the language itself.

The use of LLMs is considered for different stages of the process, analyzing when their application can add value. At the same time, the insertion of the online catalog of extensions as a component of the process is evaluated. Definition of the integration points for these resources is based on the demands identified at each stage of the process and the guidelines established for building extensions compatible with OMG standards.

6.2.5 Synthesis and Organization of the Method

Finally, the selected elements of the approaches studied, the OMG guidelines and the technological resources are organized in an integrated manner in a structured process. The

final structure of the method aims to ensure clarity, reproducibility and adherence to the technical standards of the BPMN language.

6.3 Chapter Summary

This chapter presents a methodology that seeks to investigate how LLMs can support the creation of extensions to the BPMN language, in accordance with the OMG guidelines. At first, the use of LLMs is exploratory in nature, with the aim of understanding their potential in identifying extension needs, for example in generating technical artifacts, such as the XML structure of the extension. At the same time, a process is proposed based on adapting the methods PRISE (GONCALVES *et al.*, 2020) and Braun (BRAUN, 2016) methods, aligned with the recommendations of the OMG and complemented by an online catalog of extensions developed in this research. The ultimate goal is to establish a solid basis for defining a process to support the extension of BPMN in a structured, reusable and accessible way.

7 FINAL CONSIDERATIONS AND NEXT STEPS OF THE RESEARCH

It is evident that, in recent years, the development of BPMN extensions has grown significantly. However, this process can become complex, requiring in-depth domain knowledge and experience on the part of extension developers.

From the analysis conducted in the SLR, it was verified if the reviewed studies presented a clear definition of the new extended constructs, including name, formal description, purpose and application in the proposed context. The review identified 80 studies that proposed extensions for BPMN, which were analyzed based on essential attributes, areas of application, type of extension, level of syntax and adherence to standards established by OMG.

The results revealed that 95.7% of the extensions do not preserve the original syntax of BPMN, and 55% do not offer adequate support for the modeling of new elements, hindering their practical adoption. In addition, there was a significant lack of standardization in the presentation of extensions: 55% of the studies do not provide any formal definition, 18.8% present partial definitions and only 26.3% provide complete and well-structured descriptions. These findings reinforce the need for more consistent guidelines for the definition and implementation of BPMN extensions, thus contributing to the advancement of research on the extensibility of notation.

The analysis of the results obtained with the online catalogue of extensions begins with the construct Intention of use, which presented high averages. This indicates that the participants demonstrate a strong predisposition to use the tool, which suggests interest and perception of relevance in the context of process modeling.

However, when observing the results of the Use construct, an important discrepancy is noted: the averages were significantly lower, suggesting that, despite the positive intention, the effective use has not yet materialized. This difference can be attributed to external barriers, such as lack of time, absence of institutional support or low perceived urgency at the moment of adoption of the tool.

Complementing this analysis, the high scores in the Performance Expectation and Effort Expectation constructs indicate that participants perceive the catalog as useful and easy to use, which reinforces its potential positive impact. On the other hand, the lower results in Enabling Conditions, especially in the item related to the availability of support, suggest a possible gap regarding the infrastructure or assistance needed for the continuous adoption of the tool, which may compromise its long-term use.

The SLR performed contributes to the understanding of the current panorama of the BPMN extensions, providing a comprehensive view of the field and highlighting opportunities for improvement. The creation of an accessible catalogue via web represents an important step to foster the reuse and sharing of knowledge about existing extensions, also serving as a source of inspiration for the development of new solutions. In addition, the introduction of a automated process assisted by LLMs to support the creation of extensions can represent a significant advance, offering recommendations and support aligned with good practices and standards of OMG, as well as mitigating methodological gaps observed.

7.0.1 Timeline

To continue this work, we propose the schedule of activities presented in Figure 7.1. The main planned activities are described below:

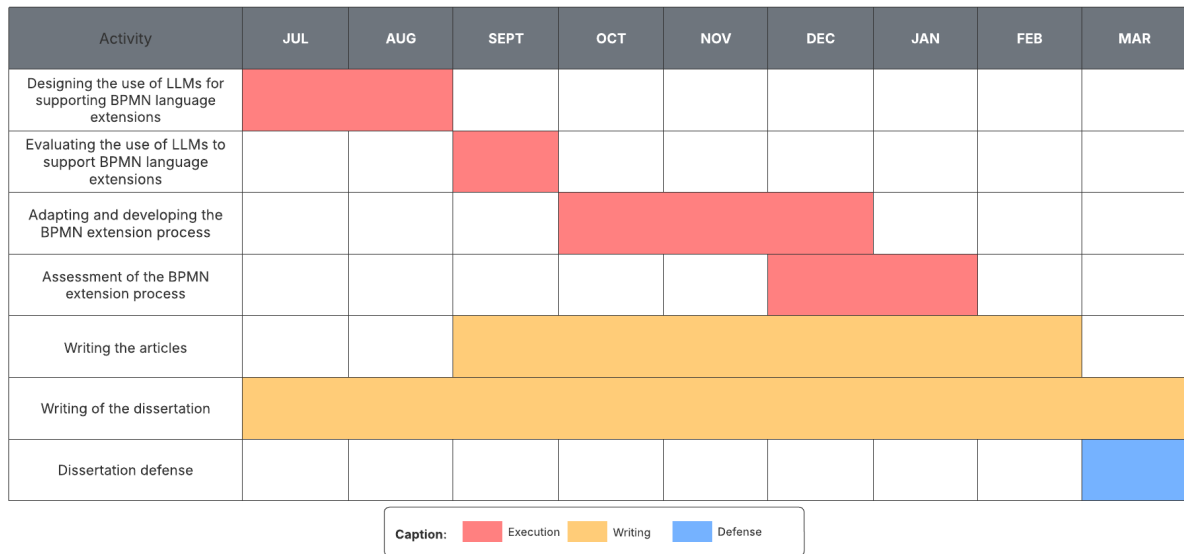


Figure 7.1 – Schedule for the completion of work

- **Designing the use of LLMs for supporting BPMN language extensions:** investigating approaches and conducting experiments to understand the potential of LLMs to support the creation and refinement of extensions to the BPMN language.
- **Evaluating the use of LLMs to support BPMN language extensions:** planning and executing evaluative tests with LLMs, analyzing their effectiveness and usefulness in assisting the development of extensions to BPMN.
- **Adapting and developing the BPMN extension process:** adapting the methods proposed by (GONCALVES *et al.*, 2020) and (BRAUN, 2016), integrating the principles and

guidelines of the OMG, with the aim of developing a structured, systematic process in line with good practices in the area.

- **Assessment of the BPMN extension process:** consolidate and evaluate the process developed, ensuring its clarity, reproducibility and applicability in different modeling contexts.
- **Writing the articles:** write and submit scientific articles to disseminate the main results of the research. The papers will cover everything from the systematic review and the extensions catalog to the experiments with LLMs and the proposed process.
- **Writing the dissertation:** write the full text of the dissertation, consolidating all the steps developed, including the execution and results of activities with LLMs and the extension process.
- **Defense of the dissertation:** presentation and defense of the work before the examining board.

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APPENDIX A – TABLE OF SEARCH STRINGS

Table 5 – Search Strings from the Database

Database	Search String	Filters
IEEE Xplore	(('BPMN' OR 'Business Process Model and Notation') AND ('Extension' OR 'Adaptation' OR 'Extended' OR 'Extends' OR 'Extensibility' OR 'Enhance' OR 'Extending' OR 'Profile'))	Period: 2019-2024
Scopus	TITLE-ABS-KEY (('BPMN' OR 'Business Process Model and Notation') AND ('Extension' OR 'Adaptation' OR 'Extended' OR 'Extends' OR 'Extensibility' OR 'Enhance' OR 'Extending' OR 'Profile')) AND PUBYEAR > 2018 AND PUBYEAR < 2024 AND (LIMIT-TO (LANGUAGE, 'English')) AND (LIMIT-TO (SUBJAREA, 'COMP'))	Period: 2019-2024 Language: English Research Area: Computer Science
Compendex (Engineering Village)	((('BPMN' OR 'Business Process Model and Notation') AND ('Extension' OR 'Adaptation' OR 'Extended' OR 'Extends' OR 'Extensibility' OR 'Enhance' OR 'Extending' OR 'Profile')))	Select the option: Subject/Title/Abstract Period: 2019-2024 Language: English
Science Direct	(('BPMN' OR 'Business Process Model and Notation') AND ('Extension' OR 'Adaptation' OR 'Extended' OR 'Extends' OR 'Profile' OR 'Enhance' OR 'Extending'))	Period: 2019-2024 Research Area: Computer Science
Springer Link	(('BPMN' OR 'Business Process Model and Notation') AND ('Extension' OR 'Adaptation' OR 'Extended' OR 'Extends' OR 'Extensibility' OR 'Enhance' OR 'Extending' OR 'Profile'))	Period: 2019-2024 Language: English Research Area: Computer Science
AIS Digital Library	(('BPMN' OR 'Business Process Model' AND 'Exten*' OR 'Enhanc*' OR 'Adapt' OR 'Profile'))	Period: 2019-2024

Database	Search String	Filters
ACM	acmdlTitle:((‘BPMN’ OR ‘Business Process Model and Notation’) AND (‘Extension’ OR ‘Adaptation’ OR ‘Extended’ OR ‘Extends’ OR ‘Extensibility’ OR ‘Enhance’ OR ‘Extending’ OR ‘Profile’)) OR keywords.author.keyword: ((‘BPMN’ OR ‘Business Process Model and Notation’) AND (‘Extension’ OR ‘Adaptation’ OR ‘Extended’ OR ‘Extends’ OR ‘Extensibility’ OR ‘Enhance’ OR ‘Extending’ OR ‘Profile’)) OR recordAbstract:((‘BPMN’ OR ‘Business Process Model and Notation’) AND (‘Extension’ OR ‘Adaptation’ OR ‘Extended’ OR ‘Extends’ OR ‘Extensibility’ OR ‘Enhance’ OR ‘Extending’ OR ‘Profile’))	Enter string in the Advanced Search op- tion Period: 2019 - 2024 Publication Type: Research paper

APPENDIX B – TABLE OF IDENTIFIED PAPERS

Table 6 – List of papers

Title	Reference
A Business Process and Data Modelling Approach to Enhance Cyber Security in Smart Cities	(HORALEK <i>et al.</i> , 2023)
A Business Process Modeling Notation Extension for Real Time Handling - Application to Novel Coronavirus (2019-nCoV) management process	(OUARHIM <i>et al.</i> , 2020)
A BPMN Extension for Business Process Outsourcing to the Cloud	(ZAROOUR <i>et al.</i> , 2020)
A BPMN extension to model inter-organizational processes in industry 4.0	(RIBEIRO <i>et al.</i> , 2021)
A Lightweight BPMN Extension for Business Process-Oriented Requirements Engineering	(INTRIGILA <i>et al.</i> , 2021)
A Model Based Framework for IoT-Aware Business Process Management	(BOCCIARELLI <i>et al.</i> , 2023)
A Migration Framework for Active BPMN Processes in Healthcare	(MANTOVANI <i>et al.</i> , 2024)
A Multi-Criteria Evaluation Approach for Selecting a Sensitive Business Process Modeling Language for Knowledge Management	(HASSEN <i>et al.</i> , 2019)
A Performance Measurement Extension for BPMN: One Step Further Quantifying Interoperability in Process Model	(HEGUY <i>et al.</i> , 2019)
A Variability-Enabling and Model-Driven Approach to Adaptive Microservice-Based Systems	(SUN <i>et al.</i> , 2021)
A Family of Experiments to generate Graphical User Interfaces from BPMN Models with Stereotypes	(DÍAZ <i>et al.</i> , 2021)
Analysis of the Runtime Resource Provisioning of BPMN Processes using Maude	(DURAN <i>et al.</i> , 2020)
BPMNE4IoT: A Framework for Modeling, Executing and Monitoring IoT-Driven Processes	(KIRIKKAYIS <i>et al.</i> , 2023b)
BPMN-E 2 : a BPMN extension for an enhanced workflow description	(RAMOS-MERINO <i>et al.</i> , 2019)
BPMN Extension for Multi-Protocol Data Orchestration	(CHIS; GHIRAN, 2022)
BPMN Extensions and Semantic Annotation in Public Administration Service Design	(ARDITO <i>et al.</i> , 2020)
BPMN Extensions for Modeling Continuous Processes	(STRUTZENBERGER <i>et al.</i> , 2021)
BPMN++: Comprehensive Business Process Modeling for Industrial Internet Application	(CHENG <i>et al.</i> , 2022)
BPMN+I to Support Decision Making in Innovation Management for Automated	(VOGEL-HEUSER <i>et al.</i> , 2020)
BPMN4MOOC: A BPMN Extension for the Design of Connectivist MOOC	(BAKKI <i>et al.</i> , 2023)
BPMN4SBP for Multi-Dimensional Modeling of Sensitive Business Processes	(HASSEN <i>et al.</i> , 2022a)
BPMN4sML: A BPMN Extension for Serverless Machine Learning	(TETZLAFF, 2022)
BPMN-L: A BPMN Extension for Modeling of Process Landscapes	(POLANCIC, 2020)
Business Process Improvement in Industry 4.0- An Interorganizational Perspective	(RIBEIRO <i>et al.</i> , 2021)
Business Process Model and Notation for Forensic-Ready Software Systems	(DAUBNER <i>et al.</i> , 2022)
Business Process Modeling Extension and Scheduling Strategy Compatible with Big Data Business	(XU; QI, 2023)
Business Process Modeling for Industrial Internet Application via BPMN Extension	(KANG <i>et al.</i> , 2024)

Title	Reference
Business Processes Meet Spatial Concerns: The sBPMN Verification Framework	(SADDEM-YAGOUBI <i>et al.</i> , 2021)
Combined Use of PBMN and Rewriting Logic for Specification and Analysis of IoT Applications	(ABBAS <i>et al.</i> , 2022)
Contributions for risk assessment of IoT-aware business processes at different granularity levels	(CARDOSO <i>et al.</i> , 2021b)
Controllability of Business Processes with Temporal Variables	(EDER <i>et al.</i> , 2019)
Designing Secure Business Processes for Blockchains with SecBPMN2BC	(KÖPKE <i>et al.</i> , 2023)
Dynamic Resource Allocation for Executable BPMN Processes Leveraging Predictive Analytics	(FALCONE <i>et al.</i> , 2024)
Evaluating BPMN Extensions for Continuous Processes Based on Use Cases and Expert Interviews	(STRUTZENBERGER <i>et al.</i> , 2024)
Extending BPM(N) to Support Face-to-Virtual (F2V) Process Modeling	(RUDAN <i>et al.</i> , 2021)
Extending BPMN Models with Sensitive Business Process Aspects	(HASSEN <i>et al.</i> , 2022b)
Extending BPMN 2.0 for intraoperative workflow modeling with IEEE 11073 SDC for description and orchestration of interoperable, networked medical devices	(NEUMANN <i>et al.</i> , 2019)
Extension of the Functional Dimension of BPMN Based on MDA Approach for Sensitive Business Processes Execution	(ALYANI, 2023)
Forensic-Ready Analysis Suite: A Tool Support for Forensic-Ready Software Systems Design	(DAUBNER <i>et al.</i> , 2024)
From Static to Dynamic Analysis and Allocation of Resources for BPMN Processes	(DURAN <i>et al.</i> , 2022)
Graphical Specification of Sensitive Business Processes	(HASSEN; GARGOURI, 2024)
How to Manage and Model Unstructured Business Processes: A Proposed List of Representational Requirements	(BUKHSI <i>et al.</i> , 2019)
Integrating IoT-Driven Events into Business Processes	(KIRIKKAYIS <i>et al.</i> , 2023a)
Integrating Quantum Computing into Workflow Modeling and Execution	(WEDER <i>et al.</i> , 2020)
IoT-Fog-Cloud Based Architecture for Smart Systems: Prototypes of Autism and COVID-19 Monitoring Systems	(KALLEL <i>et al.</i> , 2021)
IoTDM4BPMN: An IoT-Enhanced Decision Making Framework for BPMN 2.0	(KIRIKKAYIS <i>et al.</i> , 2022a)
Modeling Boundary-Spanning Business Processes in Industry 4.0 Incorporating Risk-Based Design	(RIBEIRO <i>et al.</i> , 2022)
Modeling inter-organizational business process governance in the age of collaborative networks	(RIBEIRO <i>et al.</i> , 2024)
Modeling and Execution of Blockchain-aware Business Processes	(FALAZI <i>et al.</i> , 2019)
Modeling, Executing and Monitoring IoT-Aware Processes with BPM Technology	(GALLIK <i>et al.</i> , 2022)
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